Triage from the Compressed Latent: Bandwidth-Constrained Telepathology with Delay-Tolerant Delivery

Putluru Sashidhar Reddy

Adhvik Sai Bobbasani

Deepa Nivethika

G. Dayasagar

School of Computer Science and Engineering, Vellore Institute of Technology, Chennai 600127, India

Abstract

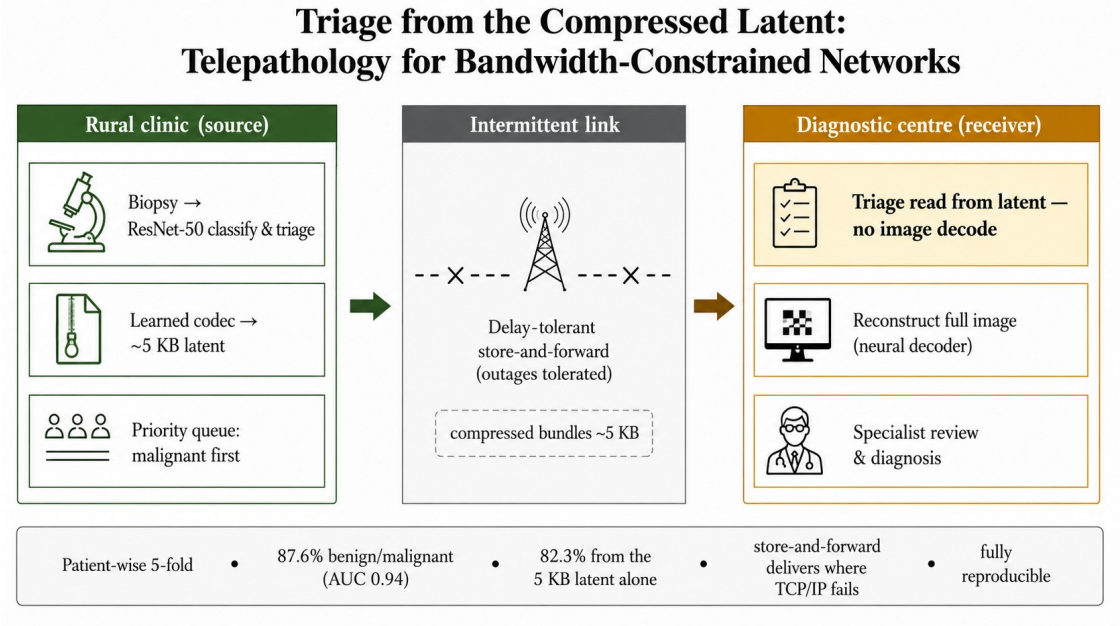
Rapid, accurate diagnosis of histopathology samples is central to the early treatment of cancer, yet rural and resource-constrained clinics frequently lack on-site pathologists and reliable connectivity. This paper shows that a learned image codec makes the compressed bitstream that a rural clinic must transmit anyway directly *triageable*: a benign/malignant triage signal—an aid to prioritization, not a standalone diagnosis—can be read from the roughly 5 KB code without reconstructing the image. We embed this diagnose-from-latent capability in an integrated telepathology pipeline that adds source-side eight-subtype classification, priority scheduling of malignant cases, and a rate-distortion-optimized codec, with delay-tolerant, store-and-forward transport—using standard single-custody forwarding, not a learned routing policy—as the layer that carries the code across intermittent links. All experiments use the public BreakHis database under the official patient-wise five-fold protocol, which prevents patient leakage. The classifier attains 87.6% (±2.2) benign/malignant accuracy with an area under the ROC curve (AUC) of 0.94, and, as a secondary capability reported for completeness rather than as a pipeline target, 49.2% (±1.5) eight-class subtype accuracy; a from-scratch baseline reaches only 29.8%. The same model reaches 95.0% under image-wise (leaky) splitting, quantifying a 45.8-point leakage inflation and underscoring why we report only patient-wise results. From the code alone, the compressed latent yields 82.3% benign/malignant accuracy without image reconstruction. Priority triage delivers urgent cases up to 31.8% faster under load with correct triage, and 28.3% at the measured latent-triage sensitivity ($r = 0.89$). In application-layer simulation, store-and-forward transport keeps delivering where connection-oriented TCP/IP collapses under severe outages. Applied without retraining to the independent BACH dataset, benign/malignant accuracy falls to 58% (AUC 0.65)—a substantial domain shift—but fine-tuning on only 100 target-site images recovers it to 81% (AUC 0.91), and the diagnose-from-latent contribution does not depend on the classifier's cross-site accuracy. We validate the compression and triage mechanisms on expert-selected biopsy fields; automated region-of-interest capture, a prerequisite for unsupervised rural deployment, is left to future work. Every reported figure is measured or reproducibly computed from released code and data.

Keywords: Telepathology; Delay-Tolerant Networking; Learned Image Compression; Histopathology Classification; BreakHis; Transfer Learning; Class Imbalance; Rural Healthcare.

Highlights

- Integrated telepathology: source-side classification, triage, learned compression, and DTN delivery.
- Patient-wise five-fold evaluation on BreakHis prevents patient-level data leakage.
- A benign/malignant triage signal (82.3%) is read directly from the ~5 KB compressed latent—a degraded-mode aid to prioritization, not a standalone diagnosis.

- In application-layer simulation, store-and-forward sustains delivery where TCP/IP collapses under outages.
- Every reported number is measured and reproducible from released code and data.



Graphical abstract.

1. Introduction

Histopathology is central to the diagnosis of cancer and many other diseases: a pathologist examines stained tissue under a microscope to identify cellular abnormalities and structural changes that indicate disease. However, expert pathologists and diagnostic laboratories are unevenly distributed, and rural or remote centres often lack local expertise. Patients must therefore travel long distances or wait for results, delaying treatment decisions in precisely those cases—such as aggressive malignancies—where early detection matters most.

Telepathology addresses this disparity by digitizing biopsy slides and transmitting them to specialists elsewhere, and it is now an established component of digital pathology practice [1], [3]. Remote review has been validated for primary diagnosis, including during the COVID-19 pandemic, with high concordance to conventional glass-slide reading [2], and it is widely recognized as a route to equitable access to expert diagnosis [1]. Yet two practical obstacles limit telepathology in low-resource settings.

First, pathology images are large. In deployment, whole-slide images can reach hundreds of megabytes or more, so efficient compression is essential to transmit them over limited links. Second, rural networks are frequently intermittent, with low bandwidth, high latency, and frequent outages. Protocols that assume continuous end-to-end connectivity perform poorly under such conditions. In addition, conventional pipelines treat all cases equally, so a clinically urgent malignant sample may be delayed behind routine examinations in a shared transmission queue.

Prior work has addressed these problems largely in isolation: computational pathology has advanced classification accuracy [5], [6], [9]; learned image compression has produced codecs that rival or exceed classical standards [13], [14]; and delay-tolerant networking (DTN) has been proposed for resilient healthcare communication in rural regions [17], [18], [19]. Comparatively little work has integrated these strands into a single, honestly evaluated telepathology pipeline.

This paper investigates such an integrated framework and evaluates it under a strict, leakage-free protocol. Our central contribution is a *diagnose-from-compression* capability: because the codec is learned, the compact bitstream that must be transmitted anyway is itself diagnostic, so the receiver obtains a benign/malignant triage decision directly from the roughly 5 KB code without reconstructing the image. Reading a decision directly from a learned compressed latent without reconstruction is itself an established idea—compressed-domain inference, or *coding for machines* [35], [36], [37]—and learned codecs have been used to compress and then classify histopathology [33], [34]. Our contribution is to bring compressed-domain inference to histopathology *triage* over a bandwidth-constrained, delay-tolerant link: the transmitted code drives priority scheduling and store-and-forward delivery, an integration that, to the best of our knowledge, has not been reported in prior telepathology work. Around this idea we build and evaluate a complete edge-to-centre pipeline:

1. **Diagnose-from-latent triage.** A lightweight classifier operating on the quantized codec latent yields an immediate benign/malignant decision (82.3%) from the compressed code alone—which a conventional codec such as JPEG cannot provide—supporting triage under extreme bandwidth scarcity (Section 5.3).
2. **Leakage-safe source-side classification.** An eight-subtype ResNet-50 + squeeze-and-excitation classifier with a derived benign/malignant decision, evaluated with the official patient-wise folds to avoid the inflated accuracies patient leakage produces [21], [22], [23] (Section 5.1).
3. **Priority-aware delivery (feasibility illustration).** A store-and-forward transport carries the learned-codec payload and transmits malignant cases first; we assess it with an application-layer simulation (Sections 5.4, 5.6) as a feasibility illustration rather than a networking benchmark.
4. **Radical reproducibility.** Every reported number is traceable to a released script and result file (Appendix A); quantities we could not measure (e.g. wire-level DTN overhead) are reported as bounds, not estimated as if measured.

Problem setting. We consider a rural clinic linked to a distant diagnostic centre by an intermittent, low-bandwidth channel that may be down for extended periods. A biopsy field must reach a specialist, and clinically urgent (malignant) cases must not wait behind routine ones. The design goals follow directly: (i) extract a usable diagnostic signal from as few transmitted bytes as possible; (ii) preserve full-image fidelity for specialist review when bandwidth allows; and (iii) guarantee eventual delivery with urgent-case prioritization despite outages.

2. Related Work

Computational pathology. Deep learning has become the dominant approach for histopathology image analysis. On the BreakHis breast-cancer database, early transfer-learning studies with residual networks reported high binary (benign/malignant) accuracy and moderate eight-subtype accuracy [5], and subsequent comparison studies systematically evaluated modern architectures across binary, four-, and eight-class settings [6]. Later methods combined deep features with gradient-boosted or handcrafted classifiers to report strong per-magnification results [7], [8], and ensemble and hybrid transfer-learning pipelines have pushed reported accuracy higher on both BreakHis and BACH [9], [10]. These works establish that transfer learning from ImageNet-pretrained backbones such as ResNet [11], often augmented with channel attention [12], is highly effective for this task. However, reported eight-class numbers vary widely with the evaluation protocol, and not all studies control for patient leakage.

Learned image compression. Classical codecs such as JPEG and JPEG2000 remain widely used, but end-to-end learned codecs optimize a rate-distortion objective directly and can surpass them. The scale-hyperprior model introduced a jointly trained prior over the latent representation [13], and combining hierarchical and autoregressive priors further improved rate-distortion performance, outperforming JPEG, JPEG2000, and BPG [14]. Channel-wise autoregressive entropy models improved efficiency while reducing serial processing [15], and comprehensive benchmarks have surveyed the field [16]. The field has since moved well beyond JPEG and JPEG2000: modern learned codecs now surpass the latest hand-engineered standard, Versatile Video Coding (VVC/VTM), on natural-image benchmarks. Transformer- and mixture-based entropy models such as

Contextformer [27] and the multi-reference MLIC family [28] report double-digit BD-rate savings over VVC on Kodak, Tecnick, and CLIC, discretized Gaussian–Laplacian–logistic mixtures [29] and wavelet-domain models [30] add further gains, and invertible-encoding transforms improve high-resolution coding [31], with Gaussian-mixture-with-attention [25] and transformer-based transform coding [26] reporting comparable gains. We compare our codec against JPEG and JPEG2000 because these remain the *clinical* storage standards for digital pathology, but we position it against this stronger backdrop and note (Section 5.2) that substituting a VVC-beating codec would only improve reconstruction, leaving our contribution—triage from the latent—unchanged. Closer to our setting, learned compression has been applied directly to histopathology: deep pathology-specific codecs outperform JPEG-XL and WebP on diagnostic-fidelity metrics [32], and autoencoder pipelines that *compress and then classify* whole-slide images [33] or drive attention-based pre-screening from neural image compression [34] show that a learned latent can carry diagnostic signal. More broadly, performing recognition directly on a learned compressed latent—without pixel reconstruction—is an established paradigm, variously called compressed-domain inference or *coding for machines*: image understanding from deep compression without decoding was demonstrated by Torfason et al. [35], and later work built joint compression–classification networks [36], adapted pixel-domain models to operate on quantized latents [37], and classified directly from variational-autoencoder codes with light fine-tuning [38]. Our contribution is therefore not the diagnose-from-latent technique itself but its application to histopathology triage over a bandwidth-constrained, delay-tolerant link: the decision is read from the quantized code that must be transmitted anyway, drives priority scheduling, and is carried by a store-and-forward layer—an integration that, to the best of our knowledge, has not been reported in prior telepathology work.

Delay-tolerant networking for healthcare. DTN uses a store-carry-forward mechanism in which data are buffered at intermediate nodes until a contact opportunity arises, enabling delivery under intermittent connectivity. Store-and-forward designs have long been proposed for rural telemedicine [17], [18], and resilient DTN-based proxy services have been demonstrated for eHealth data exchange [19]. Opportunistic and social-network DTN architectures have achieved high delivery rates for remote patient monitoring even with multi-hour delivery windows [20], and secure DTN routing has been studied for medical data in rural areas [24].

Evaluation integrity. A recurring concern in medical machine learning is data leakage: when information from the test set contaminates training—most commonly by splitting images or slices from the same subject across folds—reported performance can be dramatically inflated. Slice-level splitting has been shown to inflate accuracy by tens of percentage points relative to subject-wise splitting in brain MRI [21], feature leakage inflates AUC in mammography [22], and leakage has been shown to inflate prediction performance broadly, with small datasets most affected [23]. We therefore adopt the official patient-wise BreakHis folds throughout.

Table 1 positions this work against representative prior art. The individual strands are mature—classification [5], [6], learned compression [13], [14], DTN telemedicine [17], [19]—but, to the best of our knowledge, no prior work couples a *diagnosable* learned latent with priority-aware delay-tolerant delivery and evaluates the whole under a leakage-safe protocol.

Table 1. Positioning of this work against three representative prior-art strands. The distinguishing capability is the coupling of a diagnosable latent to resilient, priority-aware delivery; compressed-domain inference itself is established prior work (Section 2).

Capability	Computational pathology [5,6]	Learned codecs [13,14]	DTN telemedicine [17,19]	This work
Source-side subtype classification	Yes	No	No	Yes
Learned image compression	No	Yes	No	Yes
Diagnosis from the compressed latent	No	No	No	**Yes**
Priority-aware delay-tolerant delivery	No	No	Yes	Yes
Integrated edge-to-centre pipeline	No	No	Partial	Yes
Patient-wise, leakage-safe evaluation	Sometimes	n/a	n/a	Yes

3. Methodology

3.1 System Architecture

Figure 1 shows the end-to-end pipeline. At the rural clinic, a biopsy image is captured and classified; malignant cases are marked high priority. The image is compressed by a learned codec whose rate adapts to the measured link bandwidth, and the compressed bundle is queued for delay-tolerant, store-and-forward transmission. Because connectivity is intermittent, a bundle is buffered during an outage and forwarded at the next contact window. At the diagnostic centre the image is reconstructed; because the codec is learned, the receiver can additionally classify directly from the compressed latent to obtain an immediate triage decision before or without full reconstruction, after which a specialist reviews the case.

The framework targets a *region-of-interest* (ROI) telepathology workflow. Rather than transmitting an entire gigapixel slide, an on-site technician or a local screening step captures and sends the diagnostically salient fields (patches), which keeps each bundle small and matches the constrained-link setting studied here. Full whole-slide transmission would require tiling and bundle fragmentation; we return to this distinction in Section 7.

Capture adequacy — the primary failure mode of patch-based telepathology. This ROI step introduces a sampling-error risk shared by all field-selection telepathology: if a non-representative field is captured, the pipeline cannot recover regardless of the downstream AI or transport. A deployment should therefore gate capture with an automated adequacy check—for example a lightweight tissue-cellularity or focus test, or the source-side classifier's own confidence—to reject low-content or out-of-focus fields before transmission, and, where bandwidth permits, transmit several ranked regions rather than one. This failure mode is first-order, not incidental: if a fraction f of truly malignant cases are captured on non-representative (e.g. tumour-adjacent healthy) tissue, they are read as benign and never promoted, so the *effective* malignant sensitivity of the whole system degrades roughly as $r\cdot(1-f)$ regardless of how good the classifier is on well-chosen fields. Capture adequacy is therefore as important as model accuracy, and quantifying triage under realistic field-capture error is part of future work. A natural design is to let the source-side classifier itself act as the ROI selector—running sliding-window inference at the edge and transmitting only the top-K patches ranked by malignancy score—so that field selection is automated and scarce bandwidth is spent on the regions most likely to be diagnostic rather than on a manually chosen field. Because the promote-only safeguard cannot rescue a case that was never correctly sampled—a malignant slide captured on benign-looking tissue is read as benign and never promoted—robust automated ROI selection is not merely a refinement but a prerequisite for clinical deployment without an on-site pathologist, and we treat it as a primary direction for real-world use.

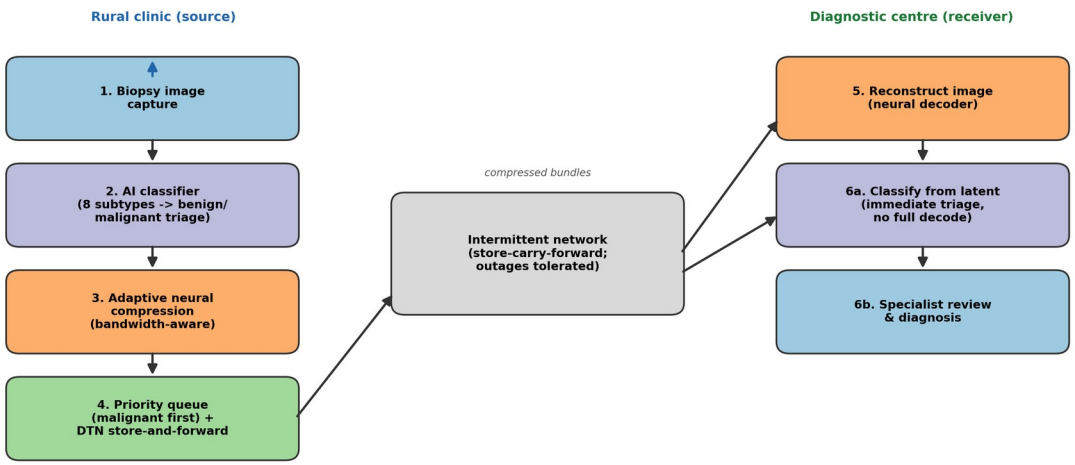


Figure 1. System architecture of the delay-tolerant telepathology pipeline. At the rural clinic (source, left) a biopsy field is (1) captured, (2) classified by the ResNet-50 + squeeze-and-excitation network into eight subtypes and reduced to a benign/malignant triage label, (3) compressed by a bandwidth-aware learned codec, and (4) queued malignant-first for delay-tolerant store-carry-forward transmission, in which a bundle is buffered during an outage and forwarded at the next contact window. At the diagnostic centre (receiver, right) the arriving code supports two paths: an immediate triage decision read directly from the compressed latent without full decoding (6a), and full neural reconstruction (5) for specialist review and confirmatory diagnosis (6b). Only the transmitted payload and the codec-side rate selection are "neural"; the routing itself is standard single-custody store-and-forward.

3.2 Histopathology Image Classification

Let I denote an input histopathology image. A convolutional network f with parameters θ predicts one of eight subtype labels, from which a benign/malignant decision is derived:

$$\hat{y} = f(I;\theta), \hat{y} \in \{1,\ldots,8\} \quad (1)$$

The backbone is a ResNet-50 [11] with a squeeze-and-excitation channel-attention block [12] and ImageNet transfer-learning initialization. The eight BreakHis subtypes comprise four benign (Adenosis, Fibroadenoma, Phyllodes Tumor, Tubular Adenoma) and four malignant (Ductal Carcinoma, Lobular Carcinoma, Mucinous Carcinoma, Papillary Carcinoma) classes; the benign/malignant label is obtained by grouping.

The dataset is highly imbalanced, so training uses an inverse-frequency class-weighted cross-entropy loss. For class c with n_c examples out of N total across C classes, the weight is:

$$w_c = \frac{N}{C \cdot n_c} \quad (2)$$

We adopt inverse-frequency loss weighting rather than resampling, based on a direct empirical comparison. Combining the weighted loss with a class-balanced oversampler over-corrected the imbalance and destabilized training, collapsing the recall of the dominant Ductal Carcinoma class toward zero; the weighted loss alone preserved majority-class recall while still lifting minority-class macro-F1. The collapse occurs because the oversampler and the loss weighting apply the same imbalance correction twice: minority subtypes are both up-sampled into most mini-batches and up-weighted in the loss, so the model can minimize training loss by predicting minority subtypes almost everywhere, and the dominant Ductal Carcinoma class—now rarely presented at its true frequency—loses recall. Correcting the imbalance once, with the weighted loss, avoids this over-correction. Focal loss is a plausible alternative that we did not tune here and leave to future work.

Classification is assessed with accuracy, precision, recall, and the F1-score (macro-averaged for the multi-class setting):

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}, Precision = \frac{TP}{TP+FP}, (3)$$

$$Recall = \frac{TP}{TP+FN}, F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} (4)$$

We additionally report the area under the ROC curve (AUC), computed one-vs-rest and macro-averaged for the eight-class task and directly for the benign/malignant decision.

3.3 Learned Image Compression

The compression model is an autoencoder with an analysis (encoder) transform E and a synthesis (decoder) transform D:

$$z = E(I), \hat{I} = D(z) (5)$$

We study two variants. A high-fidelity *residual autoencoder* stores a raw latent and reconstructs at high quality but is byte-inefficient. A rate–distortion-optimized *factorized-prior codec* (a factorized entropy model in the sense of Ballé et al. [13], [14]) couples a generalized-divisive-normalization transform with a learned entropy model and is trained to minimize a joint rate-distortion objective, where the distortion is the mean squared error (MSE) and R is the entropy-model bit-rate:

$$L = \lambda \cdot 255^2 \cdot MSE + R, R = E\left[-\log_2 p(\hat{z})\right] (6)$$

Sweeping the Lagrange multiplier λ over {0.002, 0.01, 0.05} traces a rate-distortion curve; a larger λ weights fidelity more heavily and yields a higher rate and quality. The codec uses a factorized architecture with N=128 and M=192 channels and a 14×14 latent grid, and each operating point is trained for 40 epochs at 224×224. Reconstruction quality is measured with MSE, PSNR, and SSIM:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}\left(I_i - \hat{I}_i\right)^2, PSNR = 10 \cdot \log_{10}\left(\frac{MAX^2}{MSE}\right), (7)$$

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} (8)$$

Because the codec is learned, the quantized latent $\hat{z}$ is an explicit, compact representation of the image. We exploit this by training a lightweight classifier that operates directly on $\hat{z}$, allowing the receiver to obtain a triage decision from the transmitted code without decoding the image. Because the distortion term is MSE, the codec optimizes pixel fidelity rather than perceptual or diagnostic fidelity, which tends to soften high-frequency detail such as chromatin texture (Section 5.2, Figure 7); a perceptual objective (for example LPIPS) or an adversarial term could better preserve the fine nuclear features a pathologist relies on and is a worthwhile extension.

3.4 Delay-Tolerant Transmission and Triage

Compressed images are segmented into bundles and transmitted using a store-carry-forward mechanism: during an outage a bundle is buffered and forwarded at the next contact window, so delivery survives intermittent connectivity [17], [19]. Malignant cases, identified by the source-side classifier, are placed ahead of routine cases in the transmission queue. Network behaviour is characterized by transmission success rate, throughput, and end-to-end latency:

$$SuccessRate = \frac{\#delivered bundles}{\#attempted bundles}, Throughput = \frac{data delivered}{transfer time}, Latency = T_{received} - T_{sent}$$

We refer to the proposed scheme as *Codec-Aware DTN* because it couples delay-tolerant store-carry-forward transport with the neural components of the pipeline: it carries the learned codec's compressed bundles, and the compression rate adapts to the measured link bandwidth through a threshold mapping (stronger compression when the available bandwidth is low). The routing itself is deliberately simple—a single custody copy of each bundle is buffered and forwarded at the next contact, without the message replication used by flooding schemes. The name denotes the learned-codec payload and the codec-side rate selection, *not* a learned forwarding policy. To be unambiguous: no neural network participates in any forwarding or routing decision; "neural" qualifies only the transmitted payload (the learned-codec bitstream) and the codec-side rate selection. The networking contribution is accordingly an application-layer feasibility integration—showing that store-and-forward with a single custody copy suffices to carry the diagnosable payload across intermittent links—not a new routing protocol; the novelty is the compressed-domain triage, and DTN is the vehicle that delivers it. A learned link-quality predictor—for example, an LSTM bandwidth forecaster that pre-emptively raises compression before an anticipated outage—is a natural extension, but its forecasting accuracy is not evaluated in this work and we treat it as future work.

Histopathology images are protected health information, and store-carry-forward transport buffers each bundle on intermediate custody nodes that may be outside the clinic's trust boundary. The design therefore assumes application-layer end-to-end encryption: the payload is encrypted at the clinic before the bundle is formed, so relays and couriers carry only ciphertext and never observe patient data, while the store-and-forward layer sees an opaque bundle. This is orthogonal to the routing and compression studied here—encryption adds a fixed per-bundle compute and a small ciphertext-expansion overhead at the edge—but it is a prerequisite for any real deployment, and we treat the encryption cost budget and key-management model as engineering work outside this paper's scope. Encryption is end-to-end between the two authorized endpoints—the clinic and the diagnostic centre—so the receiving server holds the key, decrypts the bundle on arrival, and then runs both the latent triage and, when needed, full reconstruction on the plaintext; intermediate relays never hold the key and carry only ciphertext. Compressed-domain triage at the receiver is therefore unaffected by encryption. Triage at an *untrusted* intermediate relay, by contrast, would require either a trusted relay or a privacy-preserving computation scheme, which we do not pursue. The bandwidth cost of encryption is negligible for this payload: an authenticated scheme such as AES-GCM adds only a fixed ~16-byte tag and a ~12-byte nonce per bundle, so a ~5 KB latent grows by well under 1% and the compression savings are preserved.

We compare Codec-Aware DTN against connection-oriented TCP/IP and two classical DTN routing strategies —Epidemic (flooding) and PRoPHET (probabilistic forwarding)—over an identical link trace, so that differences arise from protocol logic rather than from different outage sequences. All four schemes are scored under a common application-layer cost model in which the per-image transmission overhead is

$$O = (\kappa - 1)S + \rho S, \quad (10)$$

where S is the compressed payload transmitted per case—here the ≈49 KB compressed patch, and the whole compressed slide in the whole-slide analysis of Section 5.6—κ is the number of forwarded copies per bundle (1 for TCP/IP and Codec-Aware DTN, 2 for PRoPHET, and one per relay for Epidemic flooding), and ρ is a per-protocol control-overhead expressed—as a deliberate application-layer abstraction, since a real Bundle-Protocol header is instead a fixed size plus a variable payload—as a *fraction of the payload* (2% for Codec-Aware DTN, 5% for TCP/IP, 7% for PRoPHET, 10% for Epidemic; these fractions are modeling heuristics, not measured Bundle-Protocol or LTP header budgets). Two points must be stated plainly. First, the overhead scales with the *transmitted payload S*, not with the 5 KB triage latent: for a 49 KB compressed patch, $\rho \cdot S$ gives 1.0 KB for Codec-Aware DTN and 2.5 KB for TCP/IP, and the replicating schemes add whole extra copies (Table 11). Second, $\rho \cdot S$ is an application-layer abstraction, *not* a per-bundle Bundle-Protocol (BP) header: it grows with payload, whereas a real BP model adds a fixed header to *each* bundle, so wire-level overhead would instead scale with the number of bundles—for the 10,000-bundle slide of Section 5.6 that per-bundle term would dominate. Under this abstraction the differences are dominated by replication: Codec-Aware

DTN’s advantage over Epidemic and PRoPHET is forwarding a single custody copy rather than flooding duplicates, and its advantage over TCP/IP is tolerating outages without re-establishing broken sessions. Because a real per-bundle BP header can exceed a TCP/IP header, the gains reported here reflect avoided replication and outage tolerance at scale, not smaller per-bundle headers; the fractions ρ are modeling assumptions; we complement them with a wire-level simulation in The ONE (Table 12), and a full protocol-stack study across mobility regimes remains future work.

4. Experimental Setup

4.1 Dataset

Experiments use the Breast Cancer Histopathological Database (BreakHis) [4], comprising 7,909 microscopy images from 82 patients at four magnifications (40X, 100X, 200X, 400X), spanning eight subtypes (Table 2). BreakHis provides fixed-size patches of approximately 700×460 pixels, not gigapixel whole-slide images; this study therefore operates at the patch level as a proxy for the full pipeline and standardizes all images to 224×224 for the neural models. Figure 2 shows representative benign and malignant samples.

Table 2. BreakHis composition (all magnifications). The distribution is markedly imbalanced, with Ductal Carcinoma alone exceeding 40% of the images.

Subtype	
Adenosis	
Fibroadenoma	
Phyllodes Tumor	
Tubular Adenoma	
Ductal Carcinoma	
Lobular Carcinoma	
Mucinous Carcinoma	
Papillary Carcinoma	
Total	

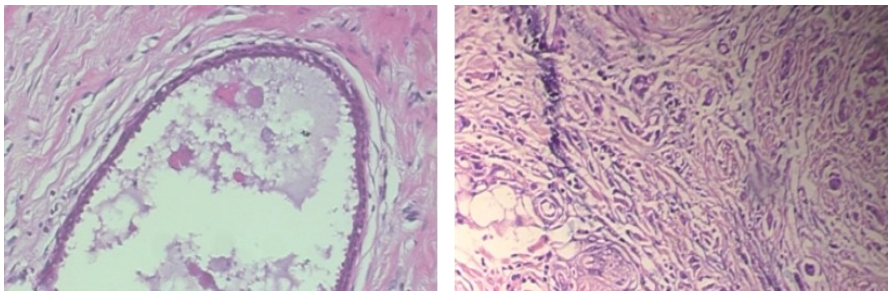


Figure 2. Representative BreakHis H&E histopathology patches, illustrating the visual task. Left: a benign case (Adenosis) with organised glandular structure and regular nuclei. Right: a malignant case (Ductal Carcinoma) with dense, disorganised cellularity and irregular, crowded nuclei. Benign versus malignant tissue is separable by architecture and nuclear morphology—the signal the triage stage exploits—whereas fine-grained discrimination among the eight subtypes is markedly harder, which is why the latent stage targets the binary decision.

4.2 Evaluation Protocol

All classification results use the official BreakHis patient-wise five-fold cross-validation, in which every patient’s images appear in either the training or the test partition of a fold, but never both. This prevents patient

leakage, which is known to inflate reported performance substantially when images or slices from the same subject are split across folds [21], [22], [23]. Metrics are reported as the mean plus or minus one standard deviation over the five folds. The pooled results use a single ResNet-50 trained on all four magnifications mixed; the per-magnification rows in Table 3 are each obtained by training and evaluating a separate model on that magnification's images under the same patient-wise folds, without ensembling across magnifications. A single mixed-magnification model is used rather than four scale-specific models so that one deployed classifier handles whatever magnification a clinic captures. Compression is evaluated on a held-out split at 224×224; the learned codec's rate is the entropy-model bit count, which is the near-optimal rate achievable by arithmetic coding, consistent with standard practice in learned compression [13], [16]. Network results are simulation outputs under explicitly stated conditions and link speeds, not measurements from a physical network.

5. Results

5.1 Classification Performance

Pooling all magnifications, the proposed model attains 49.2% (±1.5) eight-class accuracy and 87.6% (±2.2) benign/malignant accuracy, with a benign/malignant AUC of 0.939 (±0.019) and an eight-class one-vs-rest AUC of 0.739 (±0.034). Performance varies systematically with magnification: 40X yields the strongest subtype discrimination and 400X the weakest (Table 3, Figure 3). A plausible reason 400X is weakest—despite its finer nuclear detail—is that a fixed 224×224 crop at the highest magnification spans a small physical field of view, so it captures fewer cells and less of the tissue architecture (gland and stroma organization) that distinguishes subtypes, whereas lower magnifications pack more contextual structure into the same crop; the BreakHis subtypes are also often most separable at lower magnifications. We offer this as a hypothesis consistent with the observed trend, not a controlled finding. Per-class results are highly uneven and mirror the class imbalance: common subtypes such as Ductal Carcinoma reach an F1 near 0.70, whereas the rarest malignant subtypes (Papillary, Phyllodes) fall below 0.10, so the macro-F1 and the confusion matrix (Figure 4, shown at 40× where subtyping is strongest) provide a more faithful summary than accuracy alone. Reporting both the benign/malignant and eight-class levels gives a complete account of the system: the benign/malignant decision is strong and clinically actionable, while fine-grained subtyping remains difficult—consistent with the wide spread of eight-class results in the literature [5], [6], [8].

Table 3. Classification per magnification (patient-wise five-fold, mean ± SD).

Magnification
40X
100X
200X
400X
Pooled

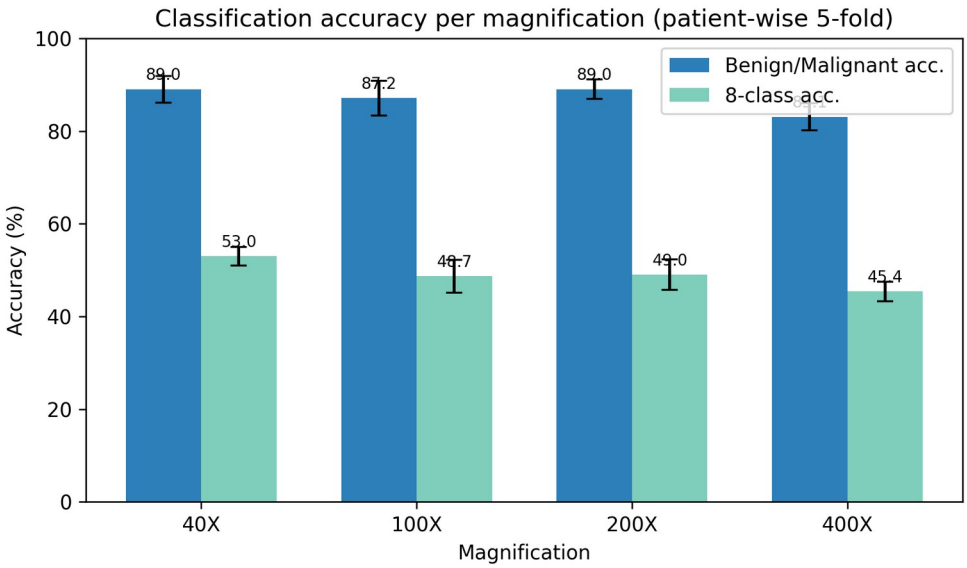


Figure 3. Classification accuracy by magnification under the patient-wise five-fold protocol; dark bars are benign/malignant accuracy, light bars are eight-class subtype accuracy, and error bars are one standard deviation across folds. Benign/malignant accuracy is robust across magnifications (83–89%), while subtype accuracy is highest at 40× (53.0%) and lowest at 400× (45.4%). The 40× advantage is consistent with a larger physical field of view per 224×224 crop, which preserves the macro-architectural context (gland and stroma organisation) that distinguishes subtypes; at 400× the same crop spans a smaller tissue area with less of that context.

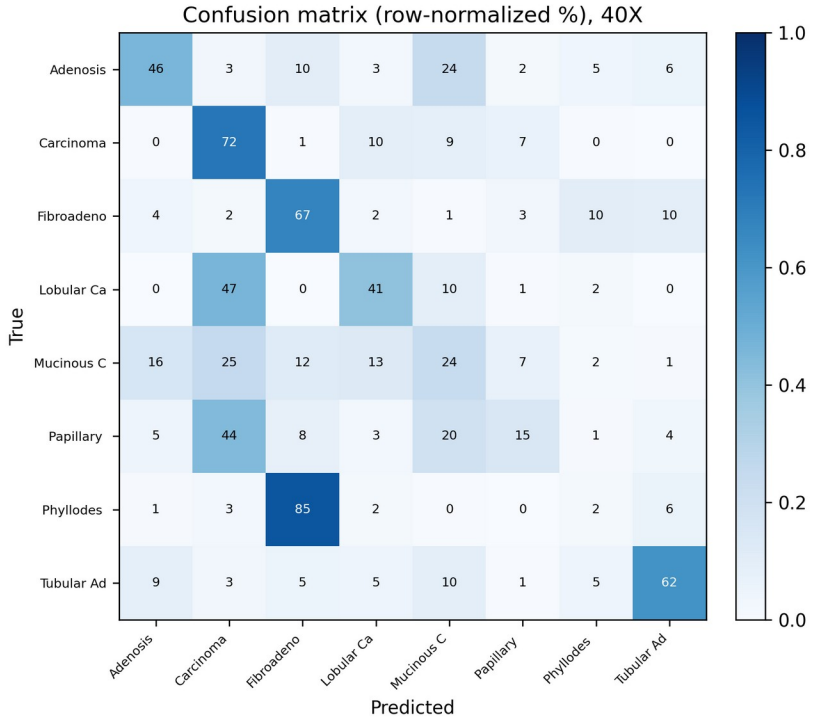


Figure 4. Row-normalised confusion matrix (%) for the eight-class subtype task at 40×; each row sums to 100% and the diagonal is per-class recall. The commonest classes are recognised well (Carcinoma 72%,

Fibroadenoma 67%, Tubular Adenoma 62%), but the rarest malignant subtypes collapse: Phyllodes tumours are almost entirely misread as Fibroadenoma (85%), Papillary carcinomas are absorbed into Carcinoma (44%), and Mucinous carcinoma is spread across several classes. This off-diagonal leakage among the rare classes is the direct source of the modest macro-F1 (0.363) and is why we report macro-F1 and the confusion structure rather than pooled accuracy alone.

Baseline comparison. To isolate the contribution of transfer learning and attention, the proposed model is compared with a plain convolutional network trained from scratch under the identical protocol (Table 4, Figure 5). The proposed model improves eight-class accuracy from 29.8% to 49.2% (a gain of 19.4 points), macro-F1 from 0.241 to 0.363, and benign/malignant accuracy from 80.8% to 87.6%. The consistent margin across every metric confirms that the architecture and transfer learning, rather than the evaluation protocol, drive the observed performance. Results are reported as mean ± SD over the five folds; the corresponding 95% confidence intervals for the pooled model are [47.4, 51.1]% (eight-class accuracy), [84.9, 90.4]% (benign/malignant accuracy) and [0.92, 0.96] (benign/malignant AUC). The eight-class interval does not overlap the from-scratch baseline's [25.1, 34.5]%, indicating a statistically reliable improvement. Paired t-tests across the five folds confirm this for every metric—eight-class accuracy ($p = 0.0002$), benign/malignant accuracy ($p = 0.008$), and macro-F1 ($p = 0.0004$); we report the parametric paired test because the Wilcoxon signed-rank test is underpowered at five folds, where its smallest attainable two-sided p is 0.0625. We also tested whether a two-stage *hierarchical* classifier (a benign/malignant stage followed by a subtype stage within the predicted group) could lift eight-class accuracy; under the identical patient-wise protocol it reached 47.7% ± 3.2 accuracy and 0.332 ± 0.025 macro-F1, which does not improve on the flat model (49.2% ± 1.5; 0.363) and lies within fold variance. The likely mechanism is error cascading: a case misclassified at the benign/malignant stage cannot be recovered by the subtype stage, and because the rarest subtypes remain data-starved within each branch, the specialization gained by splitting does not offset the errors propagated from the first stage. We did not tune a cost-sensitive first-stage threshold; lowering the benign/malignant decision threshold to raise first-stage recall (at some cost to precision) could reduce this cascade for the rare malignant subtypes and is a natural direction we leave to future work. We therefore retain the simpler flat classifier and report this negative result rather than selecting the more complex model post hoc.

Table 4. Baseline versus proposed (pooled, patient-wise five-fold).

Model	8-class Acc (%)	Macro-F1	Benign/Malignant Acc (%)
Baseline CNN (from scratch)	29.82 ± 3.80	0.241 ± 0.025	80.76 ± 3.88
Proposed (ResNet-50 + SE + transfer)	49.24 ± 1.52	0.363 ± 0.020	87.62 ± 2.22

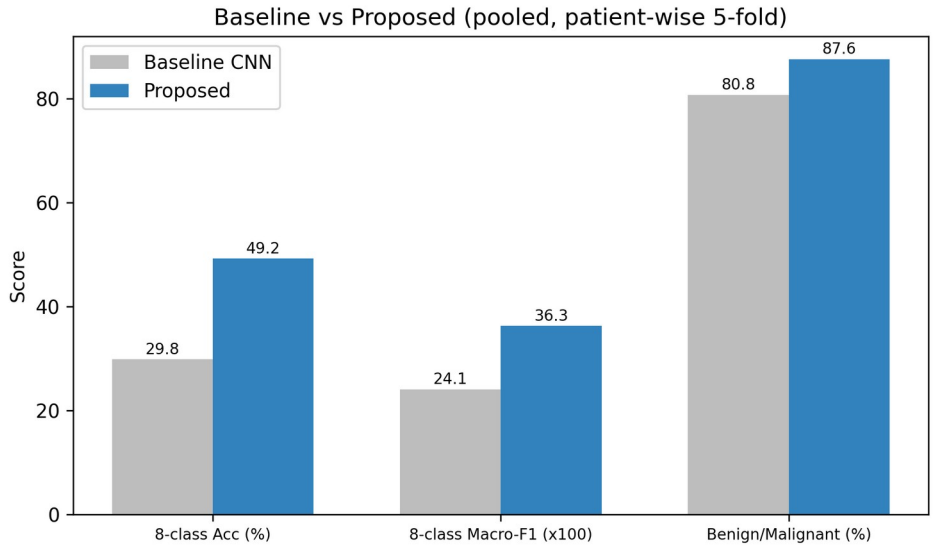


Figure 5. Proposed model versus a from-scratch baseline CNN under the identical patient-wise five-fold protocol, pooled over magnifications. Transfer learning combined with the squeeze-and-excitation ResNet-50 raises eight-class accuracy from 29.8% to 49.2%, macro-F1 (×100) from 24.1 to 36.3, and benign/malignant accuracy from 80.8% to 87.6%. The consistent margin across all three metrics confirms that the architecture and transfer learning—not the evaluation protocol—drive the improvement.

Table 5 reports per-subtype metrics pooled over the five folds and makes the difficulty explicit: the two commonest classes are recognized reasonably, while the rarest malignant subtypes collapse—the main source of the modest macro-F1.

Table 5. Per-subtype metrics (proposed model, pooled over the official five patient-wise folds). The rarest malignant subtypes (Papillary, Phyllodes) collapse, which is the main source of the modest macro-F1.

Subtype					
Ductal Carcinoma					
Adenosis					
Tubular Adenoma					
Fibroadenoma					
Lobular Carcinoma					
Mucinous Carcinoma					
Phyllodes Tumor					
Papillary Carcinoma					
Macro average					

Table 6 situates our accuracy against the published BreakHis literature and, crucially, against our own model under a leaky protocol. Reported eight-class numbers reach 90–99%, but almost all use image-wise or unspecified splits, under which a patient's images can appear in both train and test—known to inflate accuracy substantially [21], [22], [23]. We measured this effect directly: re-running the identical model under image-wise (generic stratified) five-fold splitting raises accuracy from 49.2% to 95.0% ± 0.4—a 45.8-point inflation—placing our model squarely within the range of the published leaky results. The gap to the literature is therefore a protocol artifact, not an under-powered model: 49.2% is the true, leakage-free number, while 95.0% is what a pipeline without leakage control would report. Only strict patient-wise results are comparable across studies.

Table 6. Published BreakHis eight-class accuracies versus ours. Most prior numbers use image-wise or unspecified splits, which patient/slice leakage is known to inflate [21], [22], [23]; only strictly patient-wise results are directly comparable. Shown as context, not parity.

Method	8-class Acc	Evaluation split
Ukwuoma et al. [39]	99.1%	not specified
Kolla et al. (Swin-V2) [40]	98.3%	five-fold, split unspecified
Çetin-Kaya (ensemble) [41]	96.7%	not specified
Desai et al. (ResNet-50) [43]	92.4%	augmented, split unspecified
Umer et al. (6B-Net) [42]	90.1%	not specified
Sharma et al. [44]	93.3%	patient-based aggregation, not patient-wise split
This work (image-wise, leaky)	95.0%	image-wise stratified k-fold (this study)
This work (patient-wise)	**49.2%**	**strict patient-wise (official folds)**

5.2 Compression

Compression is evaluated at 224×224 (Table 8, Figure 6). The residual autoencoder reconstructs at 36.8 dB / 0.979 SSIM but stores about 49 KB per image, which is byte-inefficient. The lightweight factorized-prior codec used for the latent-triage experiments is far smaller but trails JPEG on rate-distortion (0.86 SSIM at 4.9 KB, 0.92 at 8.4 KB). Crucially, a scale-hyperprior of the same family that we trained on BreakHis [13] is markedly stronger: it reaches 30.8 dB / 0.90 SSIM at 3.6 KB and 34.1 dB / 0.96 SSIM at 8.0 KB, surpassing JPEG on both PSNR and SSIM at matched or smaller sizes (Figure 6). A learned codec trained on histopathology therefore does beat the standard baselines here; we nonetheless use the lightweight factorized-prior codec for the diagnose-from-latent experiments. Its advantage over the hyperprior is a smaller, single-pass encoder (2.99 M parameters versus 5.07 M, with no hyper-analysis/hyper-synthesis transforms), which reduces its memory footprint. We are careful not to overstate this: neither codec dominates the edge budget—the source-side classifier (ResNet-50 + squeeze-and-excitation) is far larger than either encoder, so a device able to run the classifier can certainly run either codec. The codec-size argument is therefore about memory footprint on a constrained device, not about feasibility or encode time. Crucially, the diagnostic capability is essentially independent of which learned codec supplies the latent. We verified this directly: reading triage from the hyperprior's quantized latent under the identical patient-wise protocol yields 81.1% benign/malignant accuracy (malignant sensitivity 0.88, specificity 0.67, binary AUC 0.88), on par with—indeed marginally below—the 82.3% obtained from the lightweight factorized-prior latent (Table 9). The richer hyperprior representation therefore does not improve triage, which further supports using the cheaper factorized-prior codec at the edge. This clarifies the codec's role in the pipeline: the operational factorized-prior codec's value is not superior rate-distortion—it does not beat JPEG on fidelity (Table 8)—but that its learned latent is directly triageable, which a standard codec such as JPEG cannot offer. We measured encoder latency to test this directly (edge_latency.py). On CPU—a realistic proxy for a low-power edge node—the factorized-prior and hyperprior encoders run at essentially the same speed (13.3 ± 0.2 ms versus 13.2 ± 0.1 ms per 224×224 patch), and both are dwarfed by the ResNet-50 + SE classifier (27.1 M parameters, 58.0 ms). Two conclusions follow. First, the codec choice is effectively latency-neutral—the factorized-prior codec's benefit is its 1.7× smaller parameter footprint, not faster encoding—and because triage accuracy is codec-independent that saving carries no accuracy cost. Second, the classifier, not the codec, dominates edge compute, so reducing it is better pursued with a lighter backbone (e.g. MobileNet) than by shrinking the codec; a dedicated embedded benchmark (Jetson Nano/Raspberry Pi) would refine these desktop-CPU proxies.

This resolves an apparent tension in the rate-distortion results (Table 8): the factorized-prior codec used for the triage experiments trails JPEG on fidelity, which would hinder the specialist's confirmatory review if it were the transmitted representation. The resolution is that triage and reconstruction need not be tied to the same codec, because triage is codec-independent. For deployment we therefore recommend transmitting with the

scale-hyperprior—which surpasses JPEG on rate-distortion (Table 8, Figure 6), preserving reconstruction quality for the specialist read—and reading the triage decision from *its* latent, measured here at 81.1% benign/malignant, statistically on par with the factorized-prior codec's 82.3%. Since the two encoders are latency-equal and differ only 1.7× in memory, the factorized-prior codec is best viewed as the minimal-footprint fallback for the most constrained devices rather than the default transmission codec. The pipeline can thus deliver JPEG-beating images for the human read while retaining compressed-domain triage. Indeed, because the source classifier already dominates the edge memory budget, the factorized-prior codec's roughly 2 M-parameter saving is marginal, so on any device able to run the classifier the scale-hyperprior is the recommended default for *both* transmission and triage; the factorized-prior codec is retained only as an ablation baseline for extreme microcontroller-class nodes that run compression but not the classifier. This removes any tension around deploying a codec that trails JPEG: the recommended transmission codec (the scale-hyperprior) surpasses JPEG on fidelity, so specialist reconstruction is not compromised. Rate for both learned codecs is the entropy-model estimate (the near-optimal arithmetic-coding rate), whereas the JPEG and JPEG2000 sizes are actual bytes; the hyperprior's margin is large enough to hold under this caveat. For context, Table 7 situates our codec against recent state-of-the-art learned codecs. These report Bjøntegaard-Delta rate savings over VVC on natural-image benchmarks (Kodak/CLIC), *not* on BreakHis; the numbers are therefore not directly comparable to ours and are shown only to indicate how far the field extends beyond the JPEG/JPEG2000 clinical baselines. The takeaway is that stronger codecs exist and would improve reconstruction fidelity, but none of them provides the diagnose-from-latent capability that is this paper's contribution.

Table 7. Recent learned codecs and their reported gains. Rows above the rule are on natural-image benchmarks against VVC and are not directly comparable to our histopathology results; they establish that learned compression has moved well beyond the JPEG/JPEG2000 clinical baselines.

Codec	Reported result	Benchmark
MLIC / MLICv2 [28]	up to ~24% BD-rate saving vs VVC intra	Kodak, Tecnick, CLIC
Contextformer [27]	up to ~11% BD-rate saving vs VVC	Kodak, CLIC, Tecnick
GLLMM mixture model [29]	surpasses VVC intra (PSNR, MS-SSIM)	Kodak, Tecnick
Invertible encoding [31]	surpasses VVC, esp. high-resolution	Kodak, CLIC, Tecnick
Pathology-specific DL codec [32]	beats JPEG-XL/WebP on diagnostic fidelity	whole-slide pathology
This work (hyperprior, trained)	surpasses JPEG on PSNR and SSIM	BreakHis (histopathology)

Table 8. Rate-distortion comparison at 224×224 (averaged over 400 held-out images). Lower size at equal SSIM is better. The lightweight factorized-prior codec is listed because it is the codec used for the latent-triage experiments, not because it is state-of-the-art for compression; the scale-hyperprior is the stronger compressor and surpasses JPEG. Its lowest-rate point (0.9 KB, SSIM 0.625) is shown only to trace the extreme low-rate end of the curve and is too degraded for clinical use.

Method
Residual autoencoder (raw latent)
Factorized-prior codec (light)
Factorized-prior codec (light)
Scale-hyperprior (ours, trained)
Scale-hyperprior (ours, trained)
Scale-hyperprior (ours, trained)
JPEG

JPEG

JPEG2000

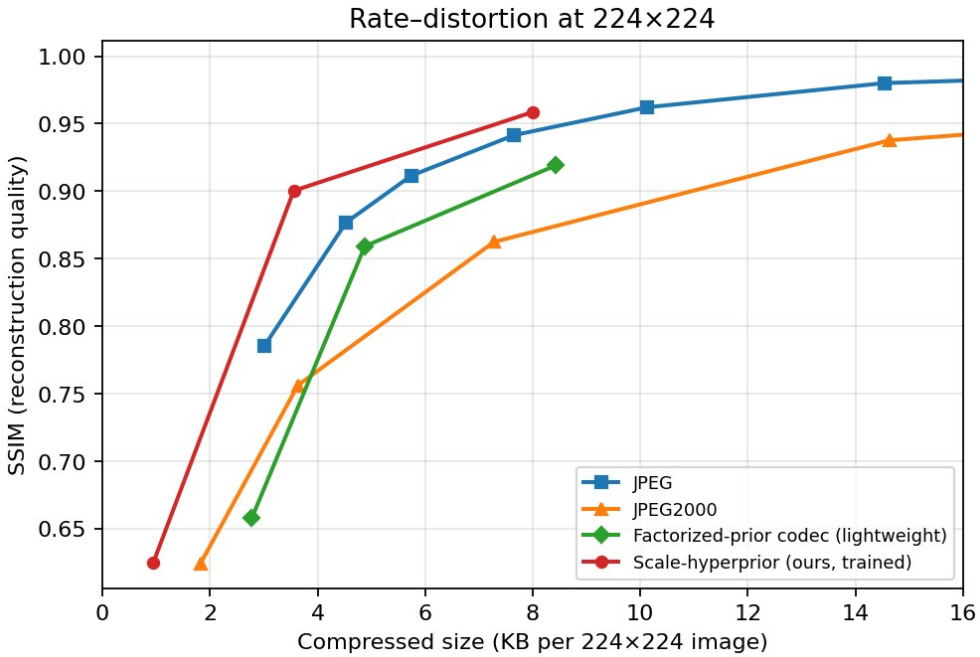


Figure 6. Rate–distortion comparison at 224×224 (reconstruction SSIM versus compressed size in KB per image, averaged over 400 held-out images; up-and-to-the-left is better). The scale-hyperprior we trained on BreakHis (red) matches or exceeds JPEG (blue) at every operating point and clearly beats JPEG2000 (orange). The lightweight factorized-prior codec (green)—the one used for the diagnose-from-latent experiments—trails JPEG on pure fidelity but is far cheaper to run at the edge. The figure establishes that learned compression is competitive with the clinical JPEG/JPEG2000 baselines, so the transmission codec need not sacrifice reconstruction quality.

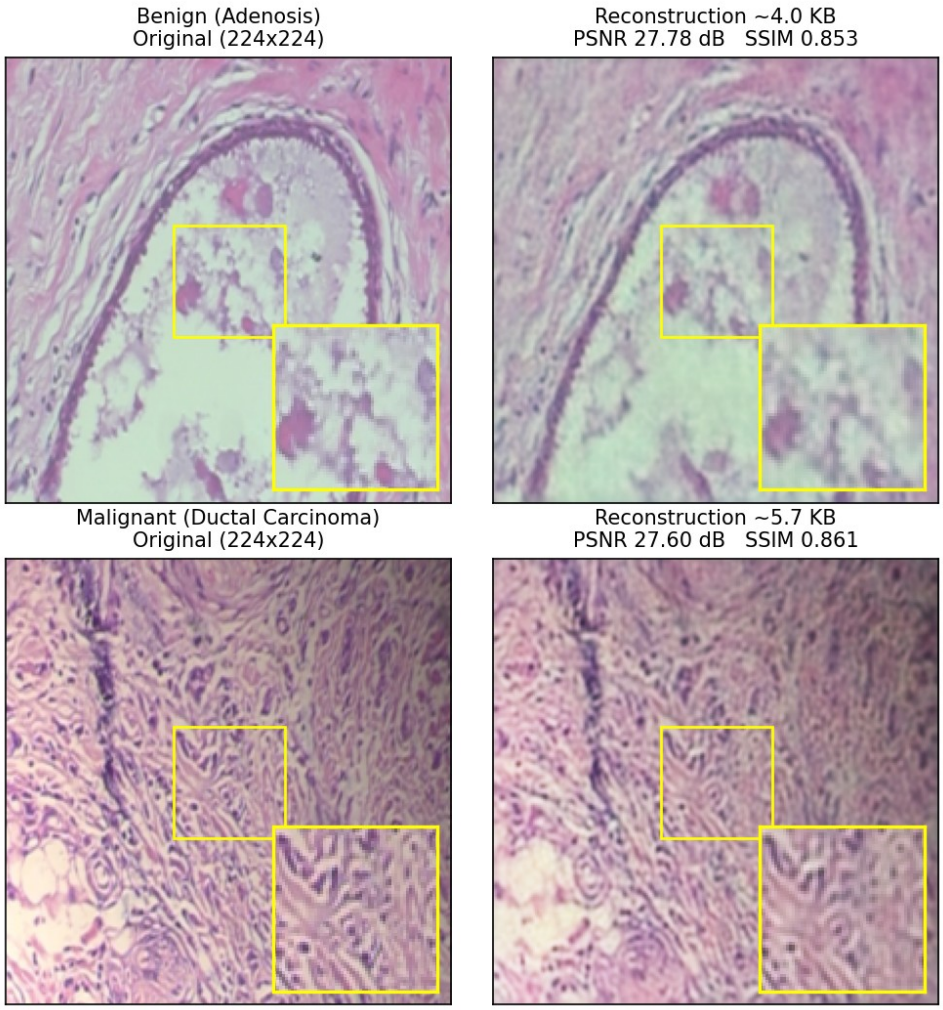


Figure 7. Qualitative reconstruction at the ~5 KB operating point (lambda = 0.01): original patch (left) versus the learned codec's reconstruction (right), for a benign (top) and a malignant (bottom) case. Per-image PSNR and SSIM are annotated, and a 4× zoom inset (yellow box) magnifies a nuclear/chromatin region so the softening can be judged directly: nuclear boundaries and tissue architecture remain visible, while the finest chromatin texture is smoothed at this rate.

5.3 Diagnosis from the Compressed Latent

A learned codec, unlike a JPEG bitstream, exposes a latent that can be classified directly—the basis of compressed-domain inference [35], [37]. We apply this paradigm to triage: a lightweight classifier trained on the codec's quantized latent (the ≈5 KB code: a 14×14×192 quantized grid of 37,632 integers that the learned factorized entropy model arithmetic-codes to ≈5 KB, because most elements are near zero and cost far less than a byte each—so the classifier reads the full 14×14×192 grid recovered by cheap entropy decoding, not a 37 KB raw array), under the same patient-wise protocol and without reconstructing the image, achieves 82.3% (±3.0) benign/malignant accuracy and 37.8% eight-class accuracy, with an eight-class one-vs-rest AUC of 0.69 (Table 9). The classifier is deliberately small and operates on the latent's native spatial grid: two 3×3 convolutional layers (192 → 256 → 128 channels) with batch normalization and ReLU, global average pooling, dropout (0.3), and a linear layer—so the triage decision is a light head over the transmitted code rather than a second full network. Because this head pools over the 14×14 grid, it deliberately collapses the latent to a

whole-patch decision and does not retain spatial localization; recovering which latent locations drive the malignant decision would need a spatially-resolved latent saliency method (for example a latent-space Grad-CAM), which is a natural interpretability extension. This is below the full-image classifier, an accuracy-for-bandwidth trade-off.

To probe what the latent encodes, we projected the pooled latent of 4,000 patches to two dimensions with UMAP (n_neighbors = 25, min_dist = 0.1, Euclidean metric, random seed 0; Figure 10). Benign and malignant patches occupy partially separated regions—a positive but modest silhouette of 0.081, consistent with the 82.3% triage accuracy—whereas magnification, a proxy for scanner and appearance variation, does not organize the space (silhouette −0.021). The latent therefore carries weak but real morphological structure rather than being dominated by an acquisition-appearance axis, which is what a triage signal read from the code requires. We caution that BreakHis is single-institution, so magnification is only a limited proxy for the stain and scanner shift that drives the BACH gap (Section 5.8); a cross-dataset latent analysis is the definitive test and remains future work.

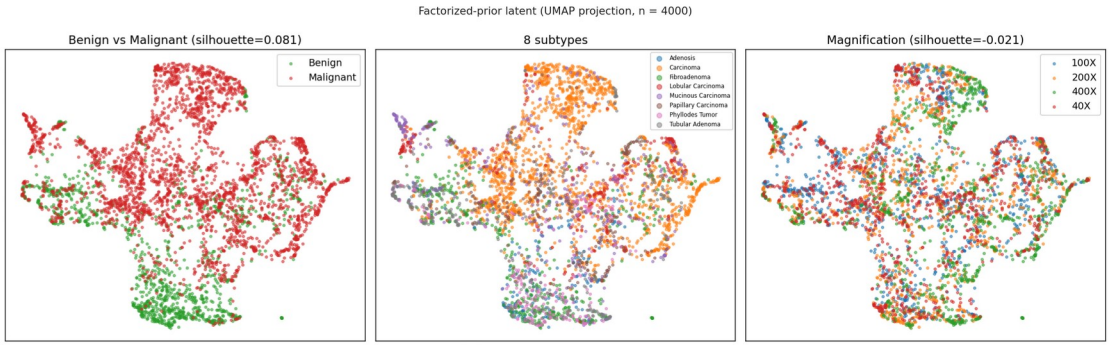


Figure 10. UMAP projection of the factorized-prior latent (global-average-pooled, 4,000 BreakHis patches; UMAP with n_neighbors = 25, min_dist = 0.1, Euclidean metric, random seed 0), colored by benign/malignant, subtype, and magnification. Benign/malignant shows modest separation (silhouette 0.081) while magnification does not organize the space (silhouette −0.021), indicating the latent encodes morphology more than acquisition appearance.

The value here lies not in avoiding the computational cost of decoding—synthesizing a ≈5 KB latent is trivial on any device—but in the fact that the diagnostic signal can be read from the transmitted code by the receiving server (or, in principle, an intermediate relay) *before* the decoder is invoked or the full image is even complete, so a case can be triaged, prioritized, and routed while the heavier synthesis transform and specialist rendering are deferred to the cases that need them. Reconstruction is performed only when a full read is required. This also clarifies that the two triage points in the pipeline are complementary, not redundant: the source-side classifier attaches a priority label that orders the *transmission* queue, whereas the receiver-side latent read lets the diagnostic centre triage, route, and begin acting on a case the moment its ≈5 KB code arrives—before the heavier synthesis decoder runs, and as an independent read rather than trusting the source's flag. Because the payload is end-to-end encrypted (Section 3.4), intermediate relays cannot read the latent, so priority decisions are taken only at the source (which holds the label) or the receiver (which holds the key); the latent read is what makes the receiver's decision immediate. Relative to decode-then-classify on an equally small JPEG, the latent path avoids a separate decoder-plus-classifier stage and does not inherit the poor reconstruction quality of a ~5 KB JPEG (SSIM ≈ 0.79 at that rate, Table 8); a controlled decode-then-classify comparison is left to future work.

The latent stage is a weak fine-grained classifier—its eight-class one-vs-rest AUC is 0.69 (Table 9), and even its benign/malignant accuracy (82.3%) sits below the full-image classifier—so it is not adequate for an autonomous screening decision. (We report the eight-class OVR AUC because a calibrated binary triage AUC on the latent was not computed; the triage's quality is better read from its 82.3% accuracy and 0.89 malignant sensitivity.) Latent triage is therefore intended strictly as a *degraded-mode fallback*, for situations severe

enough that reconstructing and reviewing the full image would otherwise take hours or days. Its role is to order the transmission queue and flag likely-malignant cases for expedited handling; every case is subsequently confirmed by full-image classification and specialist review once the image arrives. It is not proposed as a standalone diagnostic, and its clinical value lies in shortening time-to-attention under extreme bandwidth scarcity, not in replacing image review. Figure 7 shows a reconstructed patch at this operating point so that the preservation of nuclear morphology can be judged visually.

Diagnosis is robust to the codec rate. A natural question is how many transmitted bytes the triage decision actually needs. Repeating the latent classification at three codec operating points—2.8, 4.9 and 8.4 KB per image—leaves benign/malignant accuracy essentially unchanged (80.6%, 81.9% and 81.5%, all within one standard deviation of one another), against a full-image ceiling of 87.6% (Figure 8). The diagnostic signal is thus concentrated enough to survive aggressive compression: even the smallest 2.8 KB code retains the full triage accuracy, so under severe bandwidth scarcity the codec can be driven to its lowest rate without sacrificing the triage decision. This rate-invariance is a property a learned latent offers and a JPEG bitstream cannot.

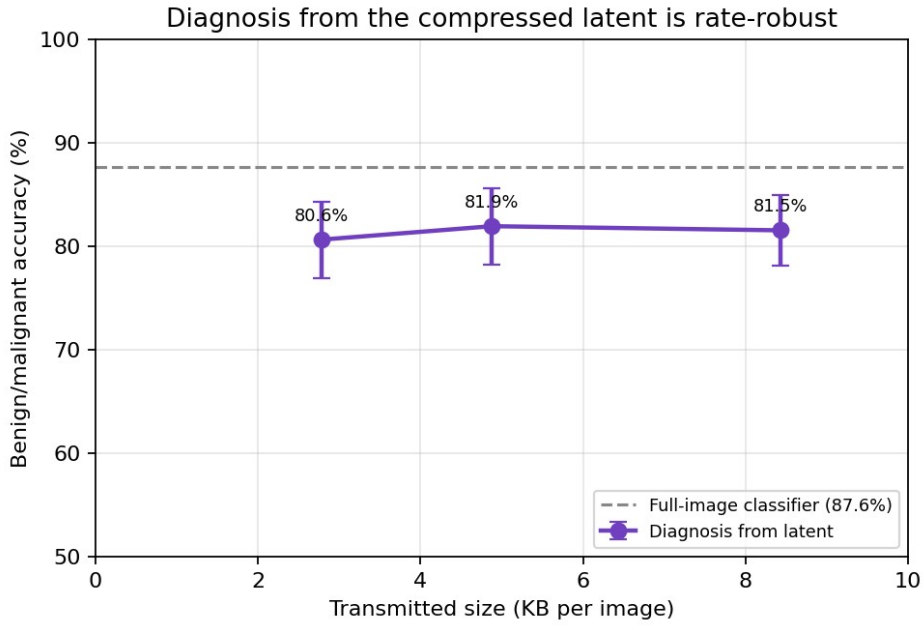


Figure 8. Benign/malignant accuracy read directly from the compressed latent, as a function of transmitted code size (error bars show fold spread). Accuracy is essentially flat—80.6%, 81.9%, 81.5% at 2.8, 4.9 and 8.4 KB—tracking a few points below the full-image classifier (87.6%, dashed line). Because even the smallest ~2.8 KB code retains the triage decision, the transmitted payload can be minimised for bandwidth-constrained links without losing triage capability: the rate can be pushed down almost freely for this use case.

Table 9. Classification from the compressed latent versus the full image (patient-wise five-fold).

Classifier input
Full image (ResNet-50)
Compressed latent (~5 KB)

5.4 Network Transmission

Transmission is evaluated across three simulated conditions on a 500 KB/s base link (Table 10). Delivery success remains 100% in all cases while throughput and latency degrade gracefully with worsening connectivity, from 374.6 KB/s and 0.14 s (Real-Time) to 111.6 KB/s and 19.48 s (Extreme). More importantly,

when a larger multi-bundle payload (the original uncompressed image) is transmitted, the connection-oriented TCP/IP protocol's delivery ratio collapses from 85% to 10.5% to 0% as outages interrupt active sessions, whereas the store-and-forward schemes (Epidemic, PRoPHET, and the proposed Codec-Aware DTN) maintain 100% delivery. This resilience under intermittent connectivity illustrates why store-and-forward transport suits the setting. We present it as an application-layer feasibility check rather than a networking contribution—the wire-level behaviour would need a protocol simulator (Section 6)—and it is directionally consistent with prior findings that DTN sustains high delivery rates in rural settings [19], [20].

Table 10. Transmission across simulated conditions, and large multi-bundle delivery ratio per protocol.

Condition
Real-Time
Delayed
Extreme

Over an identical per-image link trace, the protocols also differ markedly in transmission overhead (Table 11, Figure 9), computed with the common cost model of Section 3.4: the proposed Codec-Aware DTN incurs about 1.0 KB per image, versus 2.5 KB for TCP/IP, 52.4 KB for PRoPHET, and 200.9 KB for flooding-based Epidemic routing. The order of magnitude separating Codec-Aware DTN and TCP/IP from the replicating schemes reflects avoided duplicate copies rather than smaller headers.

We emphasize that these overhead figures are application-layer estimates produced by the cost model of Section 3.4, not wire-level measurements from a DTN testbed or an established network simulator such as NS-3 or The ONE. They deliberately omit effects that a full protocol-stack evaluation must include: on the TCP/IP side, retransmission timeouts and congestion-control back-off; on the DTN side, custody-transfer signalling, convergence-layer framing (for example LTP or TCPCL), and bundle-security blocks. The comparison should therefore be read as a theoretical application-layer abstraction of how message replication and outage tolerance differ across routing strategies—not a network-protocol validation, and not a definitive benchmark. Table 10 and Table 11 are labelled accordingly. A study in an established DTN simulator (NS-3 or The ONE) that captures wire-level Bundle-Protocol headers, custody signalling, and convergence-layer framing is planned.

Table 11. Application-layer transmission overhead per transmitted image (payload $S \approx 49$ KB compressed), as replication cost plus a per-protocol fraction ρ of the payload (Section 3.4). These are payload-fraction abstractions, not measured per-bundle Bundle-Protocol headers.

Protocol
TCP/IP
Epidemic
PRoPHET
Codec-Aware DTN (proposed)

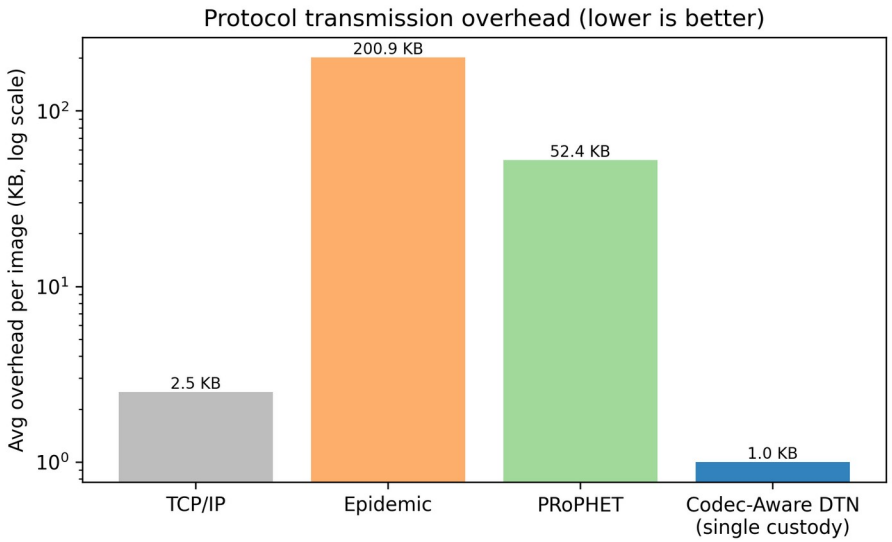


Figure 9. Application-layer transmission overhead per image by protocol (log scale; lower is better), computed from the payload-fraction model of Section 3.4. Flooding-based Epidemic incurs the most overhead (200.9 KB per image) and PRoPHET a moderate amount (52.4 KB) because both replicate bundles, whereas connection-oriented TCP/IP adds 2.5 KB and the proposed single-custody Codec-Aware DTN adds only 1.0 KB by forwarding one custody copy without replication. These are application-layer estimates, not wire-level Bundle-Protocol headers—the latter are addressed separately by the The ONE simulation in Table 12.

Wire-level validation. To move beyond the payload-fraction abstraction, we re-ran the routing comparison in The ONE, a standard delay-tolerant-network simulator that models the Bundle-Protocol layer (custody transfer, replication, buffers, and time-to-live), over a rural-courier scenario (one clinic, twenty mobile relays, and one diagnostic centre; RandomWaypoint mobility; 49 KB messages; 12 h; message TTL 720 min). The measured metrics (Table 12) confirm the abstraction's ordering: single-custody forwarding incurs the lowest replication overhead—an overhead ratio of 10.8, i.e. about 11 relay transmissions per delivered message—roughly half that of Epidemic flooding (20.0) and PRoPHET (19.9). In this connected regime all three schemes deliver 100%. The wire-level view also exposes a trade-off the payload-fraction model omits: the single custody copy raises average latency (486 s versus 92 s for Epidemic), because one bundle must be carried the whole way rather than racing many replicas. The empirical ordering supports the design choice of single custody for overhead-critical links, while making explicit that it trades latency for reduced network load. We note that the network section's contribution is not a claim of DTN routing superiority but the payload-fraction analysis showing that single-custody store-and-forward is what keeps overhead tractable once a whole-slide transfer fragments into thousands of bundles (Section 5.6); the routing comparison serves only to situate our single-custody choice among standard alternatives.

Table 12. Wire-level DTN metrics measured in The ONE (22 nodes, RandomWaypoint, 49 KB messages, 12 h). Overhead ratio = (relayed − delivered)/delivered captures real replication including per-hop relaying. Single-custody forwarding roughly halves the overhead of flooding, at the cost of higher latency—the delivery/overhead/latency trade-off that the payload-fraction model of Table 11 does not represent.

Router
Epidemic (flooding)
PRoPHET
Single custody (proposed)

5.5 Triage and Prioritization

The benefit of source-side triage is quantified directly. In batches of 20 cases contending for a single 500 KB/s link (malignant fraction approximately 0.66, matching the dataset), prioritizing malignant cases reduces their average delivery time from 1.03 s to 0.70 s, a 31.8% reduction for urgent cases, while the overall average delivery time is unchanged. Prioritization thus delivers clinically urgent samples earlier without reducing total throughput.

This 31.8% figure assumes correct triage: priority is assigned from ground-truth malignancy, so it is an upper bound on the achievable speed-up. If the latent-stage triage (0.89 malignant sensitivity) drives prioritization instead, a false-negative malignant case would receive routine priority (to be unambiguous: a false negative here means the latent stage predicted *benign* for a truly malignant case, so it is not promoted and stays in the ordinary first-in-first-out queue). To ensure the acceleration of true positives is not offset by severe delays for missed cases, the scheduler is designed to be *promote-only*: latent triage may raise a case's priority but never lower it below the default, so a latent false negative retains baseline first-in-first-out priority rather than being pushed behind all routine traffic, and full-image classification on arrival re-prioritizes any case the latent stage missed. We make this quantitative. Because a promoted malignant case receives the priority delivery time (0.70 s) while a missed one stays at the FIFO time (1.03 s), the expected malignant delivery under promote-only scheduling with latent malignant sensitivity r is $E[t] = r \cdot 0.70 + (1-r) \cdot 1.03$ s. The realistic acceleration is therefore r times the oracle value, reached fully only at $r = 1$ (Table 13). We measured the latent stage's binary confusion matrix, pooled over the five folds: it has a malignant sensitivity of $r = 0.89$ at a specificity of 0.67—an operating point that deliberately favours sensitivity, which is the right trade-off for triage, where missing a malignancy is the costly error. We state the clinical reading plainly: a malignant sensitivity of 0.89 means about one in nine malignant cases is not flagged for the fast path at the latent stage. The promote-only design guarantees such a case is never pushed below baseline priority and is re-prioritized by full-image classification on arrival, but it does not receive expedited handling. The latent triage is therefore strictly a tool for *logistical prioritization* and must never be read as a negative screen—a non-promoted case is "not yet expedited," not "benign." The realistic promote-only speed-up is therefore $r \times 31.8\% = 28.3\%$ (Table 13, measured row), close to the oracle precisely because the latent stage rarely misses malignant cases (989 false negatives against 8,161 true positives). One limit is intrinsic: promote-only cannot accelerate a case the latent stage fails to flag, so 28.3% sits below the ground-truth 31.8%, but the shortfall is small at the measured operating point. The estimate is deliberately first-order: it interpolates the two measured endpoints and does not model queue contention, so it omits the effect of *false positives*—at specificity 0.67 some benign cases are promoted and compete for the fast path. To quantify this we ran a discrete-event non-preemptive M/D/1 priority-queue simulation in which promoted cases—true positives *and* the false positives admitted at specificity 0.67—share one fast path while non-promoted cases, including missed malignancies, wait in the low-priority queue; the simulation matches the Cobham priority-waiting formula to within Monte-Carlo error. This queueing model deliberately assumes a *stable contact window* with deterministic service, so it bounds how promoted and non-promoted cases share the fast path while the link is up; it does not model outage-induced backlog—under a long outage the effective service process becomes heavy-tailed and delays accumulate beyond this bound. Outage dynamics are instead the store-and-forward layer's concern and are captured by the delivery and latency results of Tables 10 and 12. The result (Table 14) shows this false-positive risk is real but bounded: at the light-to-moderate utilisation a rural uplink operates under ($\rho \leq 0.5$), the false-positive promotions raise the mean queueing delay of a *missed* malignant case to only 1.26–1.52× the single-FIFO baseline—not catastrophic—while promoted cases are delivered faster than baseline. The degradation grows as the link saturates (2.6× at $\rho = 0.9$), the classical priority-starvation regime, so the promote-only design depends on the link not being persistently saturated. This also motivates the depromotion safeguard already present: full-image classification at the receiver re-prioritises every case, returning fast-path capacity to FIFO once a promotion is confirmed benign. A further lever narrows the false-positive load directly: because DTN traffic is bursty—a contact window can release a large backlog at once, pushing utilisation into the high-ρ starvation regime of Table 14—the scheduler can gate promotion on classifier confidence, promoting only cases above a probability threshold $\tau > 0.5$ rather than every predicted-malignant case. This trades a little sensitivity for a substantially smaller fast-path queue, while the promote-only guarantee still protects any case below τ; choosing τ against

the measured operating characteristic is a simple deployment-time control. We swept τ empirically through the latent classifier's ROC and fed each operating point into the queue model (Figure 11). At a representative load ρ = 0.6, raising τ from 0.5 to 0.95 shrinks the promoted fraction (0.71 → 0.35) and the missed-malignant queue delay (1.02 → 0.75 s), but sensitivity falls steeply (≈0.90 → 0.50; the 0.90 at τ = 0.5 is consistent with the 0.89 reported above). Because the delay is only weakly sensitive to τ at moderate load—about a 27% reduction across the whole range—while sensitivity drops by half, a low threshold (τ ≈ 0.5–0.6, retaining sensitivity ≥ 0.87) is preferable: the false-positive promotions are not the binding constraint at realistic load, so there is little reason to trade sensitivity to relieve the fast path. Raising τ is warranted only for links that sit persistently near saturation.

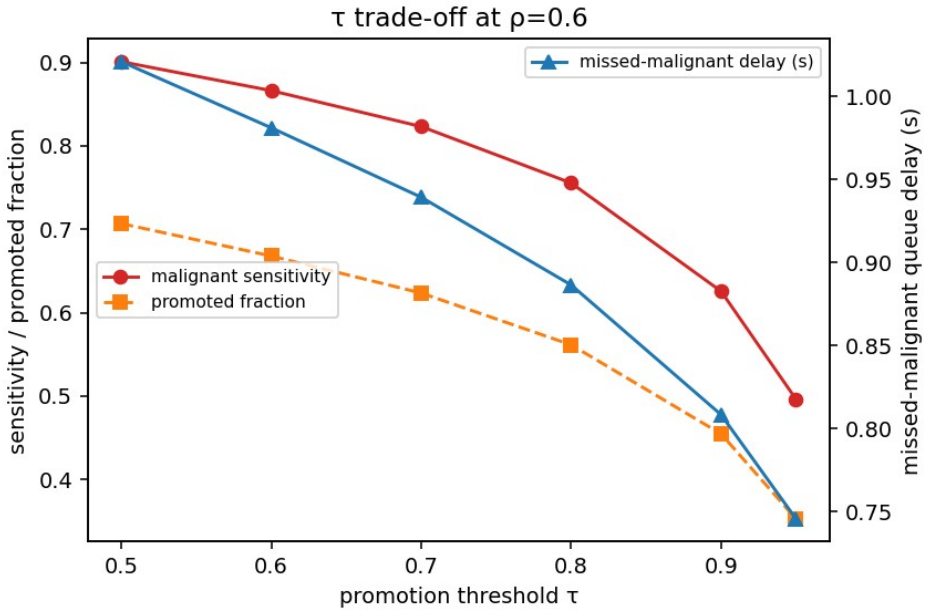


Figure 11. Empirical promotion-threshold (τ) sweep at a representative link load ρ = 0.6. Left axis: the latent stage's malignant sensitivity (red) and the promoted fraction of cases (orange, dashed); right axis: the mean queue delay of a missed-malignant case (blue). Raising τ from 0.5 to 0.95 shrinks the promoted fraction (0.71 → 0.35) and the missed-malignant delay (1.02 → 0.75 s, about 27%), but sensitivity falls steeply (0.90 → 0.50). Because the delay is only weakly sensitive to τ at moderate load while sensitivity halves, a low threshold (τ ≈ 0.5–0.6, sensitivity ≥ 0.87) is preferable: false-positive promotions are not the binding constraint at realistic load.

Table 13. Realistic urgent-case speed-up under promote-only scheduling as a function of the latent triage's malignant sensitivity r; r = 1 recovers the ground-truth oracle. Derived from the measured FIFO (1.03 s) and priority (0.70 s) malignant delivery times.

Latent malignant sensitivity *r*	Expected malignant delivery (s)	Net speed-up (%)
0.5	0.863	15.9
0.6	0.830	19.1
0.7	0.798	22.2
0.8	0.765	25.4
0.9	0.733	28.6
1.0 (oracle)	0.700	31.8
0.89 (measured, this work)	**0.736**	**28.3**

Table 14. Discrete-event M/D/1 priority-queue simulation of promote-only scheduling (latent sensitivity 0.89, specificity 0.67, so ≈70% of cases are promoted; malignant fraction 0.66; deterministic 0.78 s service per 49 KB case, 500 kb/s link). Mean queueing delay in seconds by class versus a single-FIFO baseline, cross-checked against the Cobham non-preemptive priority formula. False-positive promotions delay a missed-malignant case only modestly at low-to-moderate load (ρ ≤ 0.5) and severely only near saturation.

Link utilisation ρ	Promoted (fast path, s)	Missed malignant (FIFO, s)	Single-FIFO baseline (s)	Missed / baseline
0.3	0.15	0.21	0.17	1.26×
0.5	0.30	0.59	0.39	1.52×
0.7	0.54	1.76	0.92	1.92×
0.9	0.95	9.32	3.58	2.60×

5.6 Synthetic Whole-Slide Transmission

The patch-level results above do not stress the network the way a gigapixel whole-slide image (WSI) would. To probe that regime we simulate transmitting a synthetic 50 MB slide fragmented into 10,000 bundles of about 5 KB each—that is, the ~5 KB triage *latent* per tile, not the ~49 KB fully compressed image per tile (which would total ≈490 MB)—over the same three outage traces (Table 15). The 50 MB figure therefore models the diagnose-from-latent payload at slide scale, on which slide-level triage can proceed; full-fidelity reconstruction of every tile for specialist review would instead require the ≈490 MB of compressed images—the bandwidth–fidelity trade-off the pipeline exposes. This is again an application-layer abstraction: it models fragmentation, per-bundle control overhead, and store-and-forward delivery, but not wire-level Bundle-Protocol headers or custody bytes.

The scale exposes a difference the patch results only hinted at. Connection-oriented TCP/IP effectively fails: needing an essentially unbroken run of 10,000 successful contacts, it delivers well under 1% of the slide under every condition, because the first outage breaks the session and the slide never reassembles. All store-and-forward schemes deliver 100% by carrying bundles across outages, but their overhead—negligible at patch scale—now diverges by orders of magnitude: flooding-based Epidemic incurs roughly 205 MB of duplicate traffic and PRoPHET about 53 MB, whereas single-custody Codec-Aware DTN adds only about 1 MB. Completion time grows with outage severity, in proportion to the number of contact opportunities needed; that is the cost of resilience—the slide always arrives, but slowly. The exercise confirms that at WSI scale store-and-forward is not merely preferable but necessary, and that avoiding replication is what keeps a 10,000-bundle transfer's overhead tractable. At whole-slide scale the diagnose-from-latent stage also changes what must be sent: rather than transmitting every tile, the source-side classifier can rank tiles by malignancy score and send the top-K latents first, and the receiver aggregates the per-tile latent decisions into a slide-level triage by max- or top-K pooling, so a slide is flagged as soon as its most suspicious tiles arrive—triage becomes progressive under intermittent delivery. Robust tile ranking and slide-level aggregation are themselves future work. These overhead figures again follow the payload-fraction model of Section 3.4—Codec-Aware DTN's ≈1 MB is 2% of the 50 MB slide—so they omit the fixed per-bundle Bundle-Protocol header that a wire-level model adds to each of the 10,000 bundles; that per-bundle term would raise the real control overhead and make it scale with fragmentation, which is a further reason the store-and-forward advantage should be confirmed in a protocol simulator (Section 6). A straightforward mitigation at whole-slide scale is bundle aggregation: packing many ~5 KB tile latents into a smaller number of larger bundles sized to the link MTU amortizes the fixed per-bundle header across more payload, trading a little delivery granularity for a large reduction in the header tax, and adaptive bundle sizing to the measured contact-window capacity is a further refinement; we flag these as concrete next steps for the transport layer. The same fragmentation raises an edge-compute concern: at the measured 58 ms per tile, running the ResNet-50 classifier on all ~10,000 tiles of a slide would take on the order of ten minutes and 2–3 kJ, impractical on battery or solar power. The resolution is the tiered design standard in whole-slide pipelines: because most tiles are background or low-cellularity, a lightweight foreground detector or a small MobileNet-class relevance screen (a fraction of the ResNet's cost) first rejects empty tiles and ranks the rest, and the heavy ResNet-50 and codec run only on the top-K candidate tiles—typically hundreds, not tens of thousands. This makes the heavy model's cost proportional to diagnostically

relevant tissue rather than slide area, cutting per-slide compute and energy by one to two orders of magnitude; building and evaluating this two-tier edge cascade is a priority for the whole-slide extension.

Table 15. Application-layer simulation of a 50 MB synthetic WSI fragmented into 10,000 bundles of ~5 KB over the Real-Time/Delayed/Extreme outage traces. Overhead (replication plus control) is condition-independent in this model. TCP/IP cannot reassemble the slide once an outage breaks the session; the store-and-forward schemes all deliver, differing only in replication overhead.

Protocol
TCP/IP
Epidemic
PRoPHET
Codec-Aware DTN (proposed)

5.7 Scalability

Under concurrent-transfer stress tests (5, 10, and 20 simultaneous transmissions), all transfers complete successfully. Latency grows with concurrency but the system remains stable, indicating that the framework can scale to multiple clinics transmitting simultaneously.

5.8 External Validation

To test generalization beyond BreakHis, the trained classifier was applied—without any retraining or fine-tuning—to the independent BACH (ICIAR-2018) breast-histopathology dataset, mapping its four classes onto the same benign/malignant task—Normal and Benign → benign, In Situ and Invasive → malignant (400 images, balanced 200/200). This mapping is standard for binary screening but groups non-invasive in-situ carcinoma (e.g. DCIS, which retains an intact myoepithelial layer) with invasive carcinoma, making the malignant class morphologically heterogeneous—an additional source of the cross-dataset gap beyond scanner and stain differences. More fundamentally, BreakHis contains only benign lesions and carcinomas and has no *normal-tissue* class, whereas BACH includes Normal images: the BreakHis-trained model has never seen normal tissue and is forced to map it to benign or malignant, which destabilizes specificity beyond any purely visual shift. The gap is thus partly a class-definition mismatch, not only a scanner/stain difference. Performance drops substantially under this domain shift: benign/malignant accuracy falls from 87.6% in-domain to 58.0%, with an AUC of 0.65 against 0.94 in-domain. The degradation is strongly asymmetric—sensitivity is largely preserved (0.88, 176/200 malignant cases caught) while specificity collapses (0.28, 144/200 benign cases mis-flagged as malignant)—so the model transfers as a high-sensitivity, low-specificity screen that over-calls malignancy on out-of-distribution tissue. This bias is consistent both with the malignant-heavy BreakHis training prior (≈69% of BreakHis is malignant; Table 2) and with the appearance gap between the two datasets' scanners, stains, and magnifications. The above-chance AUC indicates the learned features retain some discriminative signal across cohorts, but the model is clearly not directly deployable at a new site without domain adaptation or threshold recalibration. Because the error is largely a prior/threshold effect, we recalibrated the decision threshold directly. Sweeping the malignancy threshold on the 400 zero-shot BACH scores to maximise balanced accuracy (the Youden-optimal operating point) lifts balanced accuracy only from 0.58 to 0.61, and it does so by flipping the model to high specificity but low sensitivity (0.88 / 0.34) at a threshold near 1.0—the model must be almost certain before calling malignant. That is the wrong direction for a triage screen, which must not miss malignancy. Threshold recalibration alone is therefore a weak fix here, bounded by the low cross-cohort ranking quality (AUC 0.65): when the ordering itself transfers poorly, no single threshold recovers much balanced accuracy. The effective remedy is therefore the 100-image fine-tuning reported below, which raises the AUC to 0.91 and the balanced accuracy to 0.81—recovering most of the in-domain performance at modest labelling cost. We note two points. First, no stain normalization (e.g. Macenko, Reinhard, or Vahadane) was applied before training the classifier or the codec, so the learned latent may partly encode BreakHis-specific stain and scanner appearance rather than pure morphology; this is a plausible driver of the BACH shift, so we tested the cheapest mitigation directly: applying Macenko stain

normalization to the BACH images before the zero-shot classifier. It did not help—balanced accuracy fell slightly (0.58 → 0.54) and AUC dropped (0.65 → 0.54)—indicating that the shift is not primarily a global colour offset that normalization can remove but reflects deeper differences in morphology, class composition, and scanner characteristics. Fine-tuning therefore remains the effective remedy, and stain-invariant training is a more promising direction than post-hoc normalization. Second, single-dataset histopathology models commonly degrade across cohorts, and the diagnose-from-latent contribution is orthogonal to the classifier's absolute cross-dataset accuracy—it applies equally to a better-generalizing backbone.

Crucially, the gap is largely one of domain adaptation rather than a fundamental failure of the learned features. Fine-tuning the same model on a small labelled BACH set—100 images (50 per class)—and evaluating on the held-out remainder (300 images) recovers benign/malignant accuracy to 81.3% and AUC to 0.91, up from 58% and 0.64 zero-shot on this held-out split (the full 400-image zero-shot in Section 5.8 gives AUC 0.65); specificity rises from 0.27 to 0.73 while sensitivity is retained (0.90) (Table 16). A modest amount of on-site labelled data therefore closes most of the cross-site gap. This raises a deployment question the rural setting makes acute: if the target clinic has no on-site pathologist, who labels the 100 adaptation images? The realistic answer keeps annotation central rather than local—the clinic uploads a small batch of *unlabelled* fields to the diagnostic centre during a connectivity window, a remote pathologist labels them once, and the updated edge model is pushed back over-the-air. Adaptation then costs one expert a few minutes per site, as a one-off per deployment, rather than requiring an on-site specialist. We flag this hub-annotation-plus-OTA-update loop as the practical form of local adaptation here, and note that fully unsupervised adaptation, with no target labels at all, remains the harder open problem. Because real target sites rarely present balanced classes, we repeated the adaptation with a benign-heavy fine-tune set (75 benign / 25 malignant, class-weighted loss, evaluated on the same balanced 250-image test). Adaptation still helps substantially over zero-shot (balanced accuracy 0.59 → 0.78, AUC 0.87), but less than the balanced 50/50 set (0.81), and the operating point skews the wrong way for triage: the benign-heavy data pulls the model toward benign, raising specificity to 0.94 but dropping malignant sensitivity to 0.61. Class-balanced or class-weighted adaptation is therefore important when the clinical priority is not missing malignancy.

Table 16. Cross-dataset benign/malignant performance on a held-out BACH split. Fine-tuning on 100 target-site images closes most of the zero-shot domain-shift gap (in-domain BreakHis AUC is 0.94 for reference).

Setting (held-out BACH test, 300 images)	Accuracy	Sensitivity	Specificity	AUC
Zero-shot (BreakHis-trained)	58.0%	0.89	0.27	0.64
Fine-tuned on 100 BACH images	81.3%	0.90	0.73	0.91

6. Discussion

The results support a clear reading. Benign-versus-malignant discrimination is strong (87.6% accuracy, AUC 0.94) and is the clinically actionable outcome on which the triage depends. Eight-class subtyping is substantially harder (49.2% accuracy, macro-F1 0.363), limited chiefly by the rarest subtypes; this is reported explicitly rather than hidden behind pooled accuracy, and the gap between benign/malignant and subtype performance is consistent with the difficulty of fine-grained subtyping reported elsewhere [6], [8]. Because the evaluation is patient-wise, these numbers reflect genuine generalization to unseen patients rather than the inflated values that leakage can produce [21], [23].

The learned codec is competitive with standard codecs and, more importantly, enables diagnosis directly from the compressed latent, coupling compression with diagnosis in a way conventional codecs cannot. On the network side, an application-layer simulation indicates store-and-forward delivery tolerates outages that stall connection-oriented transport, and priority scheduling accelerates urgent cases; we treat this as a feasibility illustration supporting the pipeline rather than a full networking result; a basic wire-level simulation in The

ONE (Table 12) confirms the overhead ordering, and a complete protocol-stack study is left to future work. Together these components form a coherent pipeline for bandwidth-limited telepathology, with diagnosis from the compressed latent as its core.

Scaling to whole-slide images. Extending the pipeline from patches to gigapixel whole-slide images (WSIs) is an architectural question we outline rather than evaluate. A WSI would be tiled into a grid of fixed-size regions; a lightweight source-side screen would rank the tiles, and only diagnostically salient regions of interest would be compressed and queued, keeping the number of transmitted units tractable. Each compressed tile maps to one or more DTN bundles, with large tiles fragmented to the convergence-layer maximum transmission unit and reassembled at the receiver by bundle identifier, while custody transfer provides hop-by-hop reliability across outages. The dominant new cost is that a single slide can expand into thousands of bundles, so per-bundle custody signalling, acknowledgements, and reassembly state grow accordingly—reinforcing why region-of-interest selection, rather than blind whole-slide transmission, is essential in this regime. Validating this design, including fragmentation limits and reassembly under repeated outages, is a priority for future work.

Threats to validity. We flag the principal threats so they can be weighed. *External validity*: applied without retraining to the independent BACH dataset, benign/malignant accuracy falls to 58% (AUC 0.65) with sensitivity preserved (0.88) but specificity reduced (0.28)—a substantial domain shift (Section 5.8). Fine-tuning on only 100 target-site images, however, recovers accuracy to 81.3% and AUC to 0.91 (Table 16), so the gap is a domain-adaptation gap rather than a fundamental one, and the classifier is deployable at a new site after light local adaptation. This is a measured limitation with a demonstrated remedy rather than an unexamined risk. *Construct validity*: the network results are application-layer simulations rather than wire-level measurements, and use patch-sized payloads as a proxy for whole slides. *Statistical validity*: with 82 patients across five folds the confidence intervals are non-trivial, so we report them explicitly (Section 5.1) rather than point estimates alone. *Internal validity*: the 31.8% triage benefit is an oracle bound; with the measured latent sensitivity ($r = 0.89$) the realistic promote-only speed-up is 28.3% (Section 5.5), so the benefit is now pinned to a measured operating point rather than assumed. None of these undermines the central diagnose-from-latent result—which is measured under patient-wise evaluation—but each marks where stronger evidence is needed.

7. Limitations and Future Work

Several limitations frame these results. First, evaluation uses patch-level BreakHis images standardized to 224×224 rather than gigapixel whole-slide images. Transmitting a few-kilobyte patch is not equivalent to transmitting a multi-gigabyte whole-slide image, which would require tiling, large-scale bundle fragmentation, and reassembly; the network latency, throughput, and overhead measured here on single small bundles will not scale linearly to that regime, where buffering, per-bundle headers, and reassembly dominate. The present framework therefore targets ROI-based telepathology—transmitting technician-selected diagnostic fields—and full whole-slide transmission is important future work. Second, the latent-space triage is a degraded-mode fallback (82.3% benign/malignant accuracy, 0.89 malignant sensitivity; the 0.69 figure represents the eight-class one-vs-rest AUC, as a binary triage AUC was not computed), not a standalone diagnostic; used autonomously it would carry an unacceptable false-negative risk, and it is deployed only to prioritize the queue ahead of confirmatory full-image review. Third, eight-class subtyping remains weak for rare classes; a two-stage hierarchical classifier did not help (47.7% vs 49.2%, Section 5.1), so more data or stronger augmentation—rather than a deeper label hierarchy—are the more promising directions. Fourth, the lightweight factorized-prior codec used for latent triage does not beat JPEG on rate-distortion; the scale-hyperprior we trained on BreakHis does surpass JPEG (Table 8, Figure 6), but is heavier to run at the edge. Matching codec quality to the edge compute budget, and reporting real arithmetic-coded rates rather than entropy-model estimates for the learned codecs, remain future work. Fifth, network results are simulation outputs under stated parameters, not measurements from a deployed rural network, and the protocol-overhead comparison rests on modeled control fractions rather than measured Bundle Protocol headers; we report a basic simulation in the DTN simulator

The ONE (Table 12) that captures custody transfer and replication, but field evaluation, as in prior DTN telemedicine deployments [17], [20], and a fuller study capturing convergence-layer and security overheads under varied mobility are still needed. Custody transfer additionally depends on a reverse channel—single-custody forwarding needs custody-acceptance signals to return to the sender so it can free its buffer—yet in a rural setting the downlink can be as intermittent as the uplink, so buffer management under asymmetric, lossy reverse channels is an open constraint; and the measured 58 ms classifier pass implies, at a typical 3–5 W edge-SoC active power, an order-of-magnitude cost of a few tenths of a joule per patch (roughly 0.2–0.3 J, dominated by the classifier, with the codec adding under 0.1 J), so on intermittent solar or battery power the classifier—not the codec—is the energy bottleneck and a lighter backbone (MobileNetV3 or EfficientNet-Lite) is the natural lever; a direct on-device Joules-per-patch measurement is still needed to confirm viability. Sixth, the adaptive-rate controller currently uses a fixed bandwidth-threshold mapping; the learned link-quality predictor outlined in Section 3.4 is not yet evaluated, and quantifying its forecasting accuracy and its effect on the rate-distortion-latency trade-off remains future work. Statistical resolution is also limited by cohort size: BreakHis has only 82 patients, so each patient-wise test fold holds roughly sixteen patients, and because one patient's images are highly correlated the effective sample size per fold is small—confidence intervals are correspondingly sensitive to a few outlier patients, and the reported standard deviations should be read in that light. The cross-site fine-tuning of Section 5.8 also used a class-balanced target set; a benign-heavy fine-tune set still improves over zero-shot but skews toward specificity at the cost of malignant sensitivity (Section 5.8), so class-balanced or class-weighted adaptation is preferable when sensitivity is the priority. Finally, PSNR and SSIM correlate only imperfectly with diagnostic fidelity in histopathology—they can penalize benign texture softening yet miss artefacts that mimic diagnostic features—so the reconstruction quality reported here is a computational proxy, not a clinical endpoint, and the automated PSNR/SSIM quality flag is likewise a reproducible signal rather than a substitute for pathologist judgement; a multi-reader multi-case study with board-certified pathologists is the required next step to establish clinical usability.

8. Conclusion

We showed that a learned codec makes the transmitted histopathology bitstream directly diagnosable: a benign/malignant triage decision is read from the roughly 5 KB compressed latent (82.3% accuracy) without reconstructing the image—a capability that a standard codec such as JPEG does not offer. We embedded this diagnose-from-latent mechanism in an integrated telepathology pipeline and evaluated it under a leakage-free protocol. The source-side classifier provides strong benign/malignant discrimination (87.6% accuracy, AUC 0.94) with an eight-subtype capability and clearly outperforms a from-scratch baseline. A lightweight learned codec compresses images to a few kilobytes and supports triage from the compressed latent (82.3% benign/malignant from the code alone), while a scale-hyperprior we trained on BreakHis surpasses JPEG on rate-distortion. Priority triage accelerates urgent cases by up to 31.8% (an oracle bound that scales with the triage stage's malignant sensitivity), and an application-layer simulation indicates store-and-forward delivery —using standard single-custody routing, not a learned forwarding policy—tolerates outages that stall TCP/IP, including for a synthetic 10,000-bundle whole-slide image. External validation exposes a real limitation—zero-shot transfer to a second dataset (BACH) is weak (58%, AUC 0.65)—but fine-tuning on only 100 target-site images restores accuracy to 81% (AUC 0.91), so the gap is closable with minimal local data. All numbers are measured or reproducibly computed from released code. The framework offers a practical, resilient pathway toward telepathology in underserved regions, with automated region-of-interest capture, cross-site domain adaptation, whole-slide scaling, and codec improvements as the main directions ahead. We are explicit that these results validate the compression and triage mechanisms on expert-selected fields; unsupervised deployment in clinics without an on-site pathologist additionally requires automated slide scanning and adequacy-gated ROI selection.

Data and Code Availability

The BreakHis database is publicly available from its original authors [4]. All source code, trained model weights, and the machine-readable result files behind every table and figure in this paper are available at the project repository (https://github.com/Dayasagar77/Triage-from-compressed-latent). Every reported number can be regenerated with the scripts listed in Appendix A.

Ethics Statement

This study used only the publicly available, de-identified BreakHis database and involved no new human-subject data collection; no additional ethics approval was required.

Appendix A. Reproducibility

Every quantitative claim traces to a script and a result file (Table 17).

Table 17. Provenance of every table and figure: the script that produced it and the result file it reads.

Table / Figure	Produced by	Result file
Table 2 (dataset)	prepare_breakhis.py (folder counts)	BreakHis_8class/
Table 3 (per-magnification)	train.py –per-mag	kfold_results_{40,100,200,400}X.json
Table 4 (baseline vs proposed)	train.py –model baseline	kfold_results_baseline.json, kfold_results.json
Table 5 (per-subtype metrics)	train.py (pooled report)	classification_report.txt
Table 6 (protocol comparison)	literature summary	references [39]-[44]
Hierarchical ablation (§5.1)	train_hierarchical.py	hierarchical_results.json
Table 8 (rate-distortion)	compare_compression.py, train_compression_v2.py	compression_comparison.json, learned_codec_results.json
Table 9 (latent classify)	classify_from_latent.py	latent_classification_results.json
Table 13 measured r	emit_latent_confusion.py	latent_confusion.json
Diagnosis-vs-bytes (§5.3)	diagnose_vs_bytes.py	diagnose_vs_bytes.json
Leakage ablation (§5.1, Table 6)	imagewise_ablation.py	imagewise_results.json
External validation (§5.8)	eval_external_bach.py	external_bach_results.json
Domain adaptation (§5.8)	finetune_bach.py	bach_finetune_results.json
Table 10 (network conditions)	benchmark_network.py	network_results_{Real-Time,Delayed,Extreme}.json
Table 11 (overhead)	benchmark_network.py	network_results.json

Table 13 (realistic triage)	realistic_triage_sim.py	realistic_triage_results.json
Table 14 (queueing sim)	queue_sim.py	queue_sim_results.json
Hyperprior-latent triage (§5.2)	triage_from_hyperprior_latent.py	hyperprior_latent_triage.json
BACH threshold recal (§5.8)	recalibrate_bach_threshold.py	bach_threshold_recal.json
Edge encoder latency (§5.2)	edge_latency.py	edge_latency.json
Latent UMAP (Fig 10, §5.3)	latent_umap.py	latent_umap.json
Imbalanced fine-tune (§5.8)	finetune_bach_imbalanced.py	bach_finetune_imbalanced.json
Figure 11, τ-sweep (§5.5)	tau_sweep.py	tau_sweep.json
Macenko stain norm (§5.8)	bach_stain_macenko.py	bach_stain_macenko.json
Table 12 (wire-level DTN)	The ONE (theone/telepathology_settings)	theone_results.json
Table 15 (synthetic WSI)	synthetic_wsi_sim.py	synthetic_wsi_results.json
Figures 3-6, 9 (results plots)	make_figures.py	results/figures/*.png
Figure 7 (reconstruction)	make_reconstruction_figure.py	fig_reconstruction.png
Triage 31.8% (Section 5.5)	benchmark_triage.py	triage_results.json

References

[1] S. Zia et al., “An update on applications of digital pathology: primary diagnosis, telepathology, education and research,” *Diagnostic Pathology*, 2025.

[2] M. Hanna et al., “Validation of a digital pathology system including remote review during the COVID-19 pandemic,” *Modern Pathology*, vol. 33, 2020.

[3] R. Weinstein et al., “Overview of telepathology, virtual microscopy, and whole slide imaging: prospects for the future,” *Human Pathology*, vol. 40, 2009.

[4] F. A. Spanhol, L. S. Oliveira, C. Petitjean, and L. Heutte, “A dataset for breast cancer histopathological image classification,” *IEEE Trans. Biomedical Engineering*, vol. 63, no. 7, pp. 1455-1462, 2016.

[5] M. Jannesari et al., “Breast cancer histopathological image classification: a deep learning approach,” *bioRxiv / IEEE BIBM*, 2018.

[6] F. Shahidi et al., “Breast cancer classification using deep learning approaches and histopathology image: a comparison study,” *IEEE Access*, vol. 8, 2020.

[7] A. Maleki et al., “Breast cancer diagnosis from histopathology images using deep neural network and XGBoost,” *Biomedical Signal Processing and Control*, vol. 86, 2023.

[8] A. Joseph et al., “Improved multi-classification of breast cancer histopathological images using handcrafted features and deep neural network,” *Intelligent Systems with Applications*, vol. 14, 2022.

[9] A. Balasubramanian et al., “Ensemble deep learning-based image classification for breast cancer subtype and invasiveness diagnosis from whole slide image histopathology,” *Cancers*, vol. 16, 2024.

[10] M. Gupta et al., “Deep transfer learning hybrid techniques for precision in breast cancer tumor histopathology classification,” *Health Information Science and Systems*, 2025.

[11] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *IEEE CVPR*, pp. 770-778, 2016.

[12] J. Hu, L. Shen, and G. Sun, “Squeeze-and-excitation networks,” in *IEEE CVPR*, pp. 7132-7141, 2018.

[13] J. Balle, D. Minnen, S. Singh, S. J. Hwang, and N. Johnston, “Variational image compression with a scale hyperprior,” in *ICLR*, 2018.

[14] D. Minnen, J. Balle, and G. Toderici, “Joint autoregressive and hierarchical priors for learned image compression,” in *NeurIPS*, 2018.

[15] D. Minnen and S. Singh, “Channel-wise autoregressive entropy models for learned image compression,” in *IEEE ICIP*, 2020.

[16] Y. Hu, W. Yang, Z. Ma, and J. Liu, “Learning end-to-end lossy image compression: a benchmark,” *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 44, 2020.

[17] J. Scholl et al., “Rural telemedicine networks using store-and-forward Voice-over-IP,” *Studies in Health Technology and Informatics*, vol. 150, 2009.

[18] E. Husni et al., “Telemedicine system based on delay tolerant network,” in *Int. Conf. on Electrical Engineering and Informatics*, 2011.

[19] E. G. Spanakis and V. Sakkalis, “Resilient healthcare communications using delay tolerant proxy services,” in *IEEE Int. Conf. on Imaging Systems and Techniques (IST)*, 2021.

[20] E. Max-Onakpoya et al., “Utilizing opportunistic social networks for remote patient monitoring in rural areas,” in *ACM SmartCitiesComm*, 2019.

[21] E. Yagis et al., “Effect of data leakage in brain MRI classification using 2D convolutional neural networks,” *Scientific Reports*, vol. 11, 2021.

[22] R. K. Samala et al., “Hazards of data leakage in machine learning: a study on classification of breast cancer using deep neural networks,” *SPIE Medical Imaging*, 2020.

[23] M. Rosenblatt et al., “Data leakage inflates prediction performance in connectome-based machine learning models,” *Nature Communications*, vol. 15, 2024.

[24] B. Prakoso et al., “Performance analysis of OLSR routing for secure medical data transmission for rural areas with delay tolerant network,” in *Int. Symp. on Electronics and Smart Devices (ISESD)*, 2016.

[25] Z. Cheng, H. Sun, M. Takeuchi, and J. Katto, “Learned image compression with discretized Gaussian mixture likelihoods and attention modules,” in *IEEE CVPR*, 2020.

[26] Y. Zhu, Y. Yang, and T. Cohen, “Transformer-based transform coding,” in *ICLR*, 2022.

[27] A. Koyuncu, H. Gao, A. Boev, G. Gaikov, E. Alshina, and E. Steinbach, “Contextformer: a transformer with spatio-channel attention for context modeling in learned image compression,” in *ECCV*, 2022.

[28] W. Jiang, J. Yang, Y. Zhai, F. Gao, and R. Wang, “MLIC: multi-reference entropy model for learned image compression,” in *ACM Multimedia*, 2023; and W. Jiang et al., “MLICv2: enhanced multi-reference entropy modeling for learned image compression,” *ACM Trans. Multimedia Comput. Commun. Appl.*, 2025.

[29] H. Fu, F. Liang, J. Lin, B. Li, M. Akbari, J. Liang, G. Zhang, D. Liu, C. Tu, and J. Han, “Learned image compression with Gaussian–Laplacian–logistic mixture model and concatenated residual modules,” *IEEE Trans. Image Processing*, vol. 30, 2021.

[30] H. Fu, J. Liang, Z. Fang, J. Han, F. Liang, and G. Zhang, “WeConvene: learned image compression with wavelet-domain convolution and entropy model,” in *ECCV*, 2024.

[31] Y. Xie, K. L. Cheng, and Q. Chen, “Enhanced invertible encoding for learned image compression,” in *ACM Multimedia*, 2021.

[32] M. Fischer et al., “Enhanced diagnostic fidelity in pathology whole-slide image compression via deep learning,” in *MICCAI Workshop on Machine Learning in Medical Imaging (MLMI)*, 2024.

[33] A. Barsi, S. C. Nayak, and S. Parida, “A deep learning-based compression and classification technique for whole-slide histopathology images,” *Int. J. Information Technology*, 2023.

[34] W. Aswolinskiy et al., “Attention-based whole-slide image compression achieves pathologist-level pre-screening of multi-organ routine histopathology biopsies,” *Modern Pathology*, 2024.

[35] R. Torfason, F. Mentzer, E. Ágústsson, M. Tschannen, R. Timofte, and L. Van Gool, “Towards image understanding from deep compression without decoding,” arXiv:1803.06131, 2018.

[36] L. Liu et al., “2C-Net: integrate image compression and classification via deep neural network,” *Multimedia Systems*, vol. 28, 2022.

[37] Z. Duan et al., “Unified architecture adaptation for compressed-domain semantic inference,” *IEEE Trans. Circuits and Systems for Video Technology*, 2023.

[38] A. Giuliano et al., “Leveraging variational-autoencoder latent representations for direct downstream classification of satellite imagery,” *IEEE Trans. Geoscience and Remote Sensing*, 2025.

[39] C. Ukwuoma et al., “Enhancing histopathological medical image classification for early cancer diagnosis using deep learning and explainable AI,” *Biomedical Signal Processing and Control*, 2025.

[40] B. Kolla et al., “An integrated approach for magnification-independent breast cancer classification,” *Biomedical Signal Processing and Control*, 2024.

[41] Y. Çetin-Kaya, “Equilibrium optimization-based ensemble CNN framework for breast cancer multiclass classification using histopathological images,” *Diagnostics*, vol. 14, 2024.

[42] M. Umer et al., “Multi-class classification of breast cancer using 6B-Net with deep feature fusion and selection,” *Journal of Personalized Medicine*, vol. 12, 2022.

[43] A. Desai et al., “Multi-class classification of breast cancer subtypes using ResNet architectures on histopathological images,” *Journal of Imaging*, 2025.

[44] S. Sharma et al., “Conventional machine learning and deep learning approach for multi-classification of breast cancer histopathology images—a comparative insight,” *Journal of Digital Imaging*, vol. 33, 2020.